
INTEGRATION OF ROBOTIC PROCESS AUTOMATION (RPA) 1.0 IN THE ACADEMIC OPERATIONAL TASK OF THE COLLEGE OF COMPUTER STUDIES (CCS): A MODULE AUTOMATING THE PROCEDURE FOR GENERATION OF WORKLOAD SCHEDULE OF FACULTY

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DEDICATION

This book is dedicated *to our loved ones, family*, who have been with us in constant support. To our friends and classmates, who are ignited with so much passion and were an echo of memorable experience. To our specials, *Trexie, Eugene, and Mernan*, who's been a hand in the most exhausting moments. And for our Team: Jaybee, Ruby, and Adhy for not just having the courage to begin but also having the strength to accomplish a challenge meant to be faced. This milestone will serve as a reminder that we are capable of doing more than we ever imagined.

Team Boskoop

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ABSTRACT

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Employees in modern workplaces increasingly rely on technology to manage operational processes yet requires proper planning and implementation to remain efficient. In the College of Computer Studies (CCS) at Camarines Sur Polytechnic Colleges, several long-standing procedures particularly in the manual preparation of documents have become repetitive, and time-consuming. This study, The Integration of Robotic Process Automation (RPA) to the Academic Operational Tasks of the College of Computer Studies, aims to enhance the Generation of Workload Schedule of the Faculty by assessing its automation potential and effectiveness. A multi-method approach was used to meet the study's objectives. Qualitative methods, including interviews and observations, identified workflow patterns and faculty concerns, while quantitative methods provided numerical interpretations through scoring formulas, data inspection, and simulation. Guided by the Business Process Management (BPM) Framework, the researchers (1) analyzed the existing workflow mapped in As-Is BPMN model, (2) evaluated automatability using Multi-Criteria Scoring Model and Value-Complexity Matrix, and (3) developed a To-Be Model. Nineteen tasks were identified, with seventeen (89.47%) classified as highly automatable and two (10.53%) as medium. Four tasks were deemed highly complex due to complex plotting requirements. The redesigned automated workflow implemented Microsoft Power Automate Desktop in VirtualBox showing a reduction in processing time from an average of 4 minutes and 60 seconds manually to 37 seconds with automation, achieving 100% accuracy across ten generated documents. The study concludes that RPA 1.0 significantly improves efficiency and reliability and recommends exploring more advanced RPA versions, to other departments, and refining workflows for future research.

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CHAPTER 1

INTRODUCTION

This chapter provides an overview of this research including the project context, purpose and description, scope and delimitation, and project dictionary.

1.1 Project Context

Technological advancement is believed to have enhanced the basics of daily activities, which, in our current time, has helped manage modern work-related operations. Manual work was eased by applying tech supported measures to easily manage tasks that were once inefficient [3]. Although technology may have been seen to be beneficial rather than doing the primitive ones, empirical study shows that it may also contribute to inefficiencies often associated with an outdated working environment, high volume of tasks, and inability to adapt to some working fields [12]. According from a study of Jomuad et al. [2021], work experience and burnout often stems from work burdens [5] that are sometimes minimal but are critical once added or piled up.

Educational institutions like the College of Computer Studies at the Camarines Sur Polytechnic Colleges operate technology-driven approaches to its operations. In fact, in its previous years, the college has deployed Robotic Process Automation (RPA) in enrollment tasks that were found functional in the college. Time durations were cut down in automating procedures that require heavy uptime work like encoding grades and obtaining class lists [4]. Moreover, the integration of RPA solutions posed a great opportunity to refine manually processed tasks and re-digitalized repeated action of activities.

In spite of this milestone, the college still faces burdens on its operational tasks particularly in creating workload schedules. Studies show that timetable scheduling requires significant time and meticulous planning [6]. Although the role of technology were considered to ease manual operated tasks, the tasks of plotting these schedules in each faculty show significant burden to the functionality of both the administrative office and the instructors. Currently, the tasks of plotting

each schedule were passed down to each instructor to process documents which can result in inefficiencies and work complaints. Hence, a move to integrate RPA shows a relevancy in terms to effectively managing a complex work and satisfaction to work-related tasks [9].

Considering that the process of generating a workload schedule of each faculty amounts to a huge manual-processing time, utilizing the use of automation technology, according to Charles et. al, offers a strategic move to ease and efficiently optimize workflow of employee tasks [5]. Automation technology interacts well with applications and executes ‘flows’ the same as how a user typically performs a task repetitively. This allows users to deploy robots to interact within applications thus supporting their tasks in certain processes that are still needed for human interaction [2, 7]

Furthermore, automation is becoming an asset that results in a strategic nature and not a tactical resource [1]. The use of this technology significantly changed the performance of academic institutions like the College of Computer Studies of the Camarines Sur Polytechnic Colleges whereas the digitalization of the current manual operations has provided a positive performance that aid efforts in time-consuming loop of tasks [4].

1.2 Purpose and Description of the Project

The generation of workload schedule of the faculty is an essential administrative task in the institution. Currently, this process has been carried out manually, requiring faculty to spend significant time in inputting and validating the data. A decision to integrate RPA suggests that integrating RPA optimizes work-related task [11].

The primary goal of this study is to automate the process of plotting faculty workload by integrating Robotic Process Automation (RPA) in the College of Computer Studies (CCS). Although local studies implemented automation in the college, the purpose of integrating RPA specifically in the faculty workload generation offers a significant step in the current level of

operational tasks. As a result, the CCS Faculty benefit from generating workload reports to improve efficiency and reduce human errors.

The use of software robots in RPA relied on the interaction of the user that effectively triggered the automation to start the scripted workflows. The interaction of the user represents the initial phase of automation, while the performance of software bots on rule-based and repetitive tasks indicates the execution phase. Automation can be a leap of towards operational capability in providing the faculty the reliance to auto generate scheduling plans in the execution of RPA to the college.

1.3 Objectives of the Project

The general objective of this study is to improve the current process of faculty workload generation of the College of Computer Studies by integrating Robotic Process Automation (RPA 1.0).

Specifically, this study aims to achieve the following:

1. Examine the current process of generating the faculty workload schedule and assess its automatability.
2. Integrate RPA 1.0 on the automatable tasks of faculty workload schedule generation.
3. Evaluate the accuracy and efficiency of the implemented automation of faculty workload schedule generation.

1.4 Significance of the Study

This section discusses the significance of Integrating RPA in the plotting of the faculty schedules of the College of Computer Studies emphasizing its potential contributions to the institution and highlighting the value of RPA technology. The researchers believed that it would be beneficial to the following sector:

Faculty. Automating the generation of workload schedule will help enhance operational efficiency by providing better productivity which allocates more time in teaching priorities. The integration of RPA to the workload schedule will also improve work efficiency of the faculty.

College of Computer Studies (CCS). This study will help improve the operational efficiency of the faculty by adapting RPA in plotting the schedules of the faculty. This will also allow the college to reinforce commitment through research development and progressive performance.

Camarines Sur Polytechnic Colleges (CSPC). This will contribute to the goal of the institution in strengthening knowledge of innovative technology and transcends academic progress.

Researcher. This will help improve the researcher's knowledge about Robotic Process Automation, particularly on how it can be used in operational tasks such in administrative settings.

Future Researcher. This study may serve as a guide for future researcher to explore and conduct studies related to RPA that may also provide an insight to further improve the results of this study.

1.5 Scope and Limitations of the Project

The development of this study focused on examining key processes related to the plotting of schedules done in the College of Computer Studies. Using research instruments such as Automatability Scoring Sheet and Multi-Criteria Scoring Model, the researchers identified automatable tasks based on the manual practices in the current setting. The data collection method used semi-structured interviews and observations to understand the involvement of the schedules provided in the Summary of the Subject Load Notice (SLN) file, while the integration of software robots from RPA 1.0 using Microsoft Power Automate Desktop were simulated and were inspected based from its ability to plot data from Microsoft Excel.

Additionally, this study will not include other operational processes in the college and will only deploy RPA 1.0 on the identified process. The basis of plotting each schedule will rely on the indicated data extracted from the Subject Load Notice (SLN) file, additional schedules that weren't indicated (e.g Overload, Quasi-teaching, Research Extensions, etc.) are excluded since it relies on

complex decisions and human intervention not covered by RPA 1.0. Furthermore, the study's findings won't apply to all 63 faculty members and only a number of participants will be sampled in achieving the desired results. This study is limited in identifying the effectiveness of integrating RPA in future projects of the college. These limitations were set to consider the proponents' capabilities in achieving the desired results and output.

1.6 Project Dictionary

To provide better understanding and to avoid problems in understanding the terms used, the following technical terms are conceptually and operationally defined.

Accuracy. The quality of being precise and exact [13]. In this study, accuracy shows how closely a result or output matches with the accepted value or expected output.

Automation. The use of technology to perform task with minimal human interactions in order to improve process consistency, efficiency, and reliability [8]. In this study, automation is a tool integrated with Power Automate Desktop that launches software robots to its process.

Model. Is a simplified representation of a system at some particular point in time or space intended to promote understanding [14]. In this study, the model will serve as the blueprint guide for creating the As-Is and To-Be processes of the college.

Robotic Process Automation (RPA) 1.0. A technology that uses software robots to automate repetitive task by copying human interactions with digital systems [10]. In this study, RPA is a technology used to automate the process of generating the workload schedule of the faculty in the College of Computer Studies.

Process. Process is a series of actions that produce something or that lead to a particular result [15]. In this study it is the step-by-step procedure done by the employee in the college.

Software Robots. Software robots do repetitive and lower value of work, like logging into applications and systems, moving files and folders, extracting, copying, and inserting data, and

filling in forms [16]. In this study it is the automated agents that efficiently perform the repetitive and rule-based tasks involved in generating faculty workload schedules.

Tasks. Tasks is a library for controlling task execution during a Robot Framework run [17]. In this study, it is a particular series or set of actions that are performed manually by the employee.

Time-Efficiency. A broad concept that includes philosophical, ontological, and epistemological perspectives. It is used as a strategic tool to connect current operations with future development to improve decision making, competitiveness, and managerial effectiveness [8]. In this study it is the average amount of time calculated in the generation of workload schedule without RPA and the ability to generate the expected outcomes with RPA.

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CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter provides an overview of existing literatures and studies related to workload schedule generation using Robotic Process Automation (RPA). This includes key topics such as process of generating workload schedule, implementation automation in education sector, framework/methodology for RPA implementation, assessing accuracy and efficiency of implemented RPA and impact of RPA 1.0 in generation of workload. This also includes the synthesis of the state-of-the-art and the gap bridge of the study.

2.1 Related Literature and Systems

To gain comprehensive overview related to the study, the researchers examined previous studies and highlighted the important findings relevant to the course. These were discussed thoroughly.

2.1.1 Process of Generating Workload Schedules

Operational Excellence comes from different aspects, one of such is the effective workload scheduling. Organizations can improve their management and fluently run their business operations by having a well-planned schedule. Effective workload scheduling enhances productivity, minimizes delays, and ensures the optimal use of resources across various sectors.

According to the study of Jomud et al. [2021] proper work assignment and work distribution are essential to maintaining instructors' efficiency and reducing stress. The researchers emphasize the importance of school administrators to balance the work schedule of teachers in public schools where the study revealed that faculty do frequently experience high relative amount of burnout because of work overload [13] it is why adhering to proper workload assignment can avoid stress and burnout. But accurate and efficient timetable scheduling requires significant time and meticulous planning [14]. Many employees struggle in compliance to regulatory requirements, and since timetable scheduling is one complex work, it underlies complicated task production.

In the study of Vikulin et al [2020], the researchers developed an algorithm for the formation of the training load using automation. The algorithm were considered to understand the application of automation in a part of the work that includes the important operation in timetable scheduling such as processing initial data, combining the area of study, calculating the teacher's working hour and assigning teachers. The use of automation in the timetable schedule process is improved by the automation for training load formation, issues like last-minute changes and teacher preferences may still affect that's why the automation need to depend on structured tasks to execute its efficiency.

In the context of automated system, an innovative Decision Support system (DSS) which was made by Xue et al. [2024] was designed to address the University Course Timetabling Problem (UCTP) [24]. This provides an organized approach for addressing the inherent complexity of the UCTP process. According to the researchers, creating a timetable is also like making work schedules that are difficult to process and needs attention to detail. This potentially disrupts management capabilities that often stems to inefficient practices. The results showed that by integrating a DSS into UTCP, the researchers were enabled to provide faster timetable creation minimizing manual errors, while making it an essential tool for academic institutions looking to improve their scheduling operations.

During the study of Arunachalam et al. [2024], it was discussed that the problem of developing schedules for colleges and schools has been explored in the past using a variety of methods [3]. According to the researchers, producing time-table is easy if an automatic time-table generator is present. This means that with the recent studies in the past, the researchers were able to produce a time-table or so-called workload schedule (in this study) to produce potential schedules for the working days. In addition, a feasible timetable algorithm has manage both hard and soft limitation which was developed by the researchers. This method were found to be an

essential approach that helps to overcome the challenges of creating a time schedules for the educators.

As illustrated from the study of Vikulin et al. [2020], the development of a timetable generation algorithm is a major advancement in the field of educational scheduling [14], such automated system made by Arunachalam et al., gave importance in providing lasting benefits in the workplace. Furthermore, other relevant sources like in th study of Aravinth et al. [2022] noted that the shift toward digitization has increased the demand for automation technology in business operations. These factors has showed to lower the dependency on human-centric tasks on workforce which provided a vast capability of putting RPA as an assistant tool for a business flow. The work schedule that were found to carry out automation without the intervention of a human being [2], has influenced an innovative design to potentially apply automation in the education sector.

The role in technology focusing in the adaption has emerged, enabling organizations to perform activities that were once repetitive and rule-based. The use of technology has helped perform various tasks without direct human intervention that enabled work management effective in applying appropriate measures. While many automated-designed systems were used to perform the process of generating workload schedules, the studies added that the shift of digitization not only improves operational efficiency, but it also allows the employees' time for more strategic and value-driven activities which can be used in technology driven institutions.

2.1.2 Implementing Process Automation in the Education Sector

As technology increasingly gains significance in role and utility, it is evident that it has profoundly influenced perspectives and reshaped global conventions. Consequently, organizational sectors, particularly in education, have embraced technological advancements by implementing automation to enhance operations that are time-consuming and repetitive in administrative activities.

The manual operations of work-related tasks require such employee resources which occasionally render some procedures ineffective. As discussed by Zahurin et al. [2024], heavy workloads result in significant levels of stress and a reduction in the quality of their teaching [25]. This could only mean that their overall performance in properly executing operational process in conducting such repetitive work-related tasks requires on-hand skills. In the same study, the researchers discussed that a possible solution method to optimize teachers' workload will incorporate Robotic Process Automation (RPA). Additionally, Technology like RPA can assist teachers in adapting to growing demands and significantly impact their teaching efficiency [25].

With these reasons, integration of automation can possibly enhance productivity in many areas and enables them to adapt to unanticipated market changes. Administrative and faculty workload management are essential because they boost task productivity in areas that are most needed. Zahurin et al., added that teachers can use RPA to better fulfill their primary goal of teaching students and thereby increase efficiency, accuracy, productivity, and flexibility.

Several scholars have reflected on the effective applications of RPA in learning institutions, projecting the revolutionary impact of RPA on student service, enrollment, administrative processes, and institution effectiveness. In a journal by Selamat et al. [2024], the researchers concluded that some studies have begun to explore the possibility to deploy RPA based in real world experiences [21]. Arguably, it has been discussed that RPA might meet changing demands, remain ahead of the curve when used for adaptively changing educational landscapes.

Moreover, allowing these practices effectively can maximize its long-term benefits and address potential challenges in educational institutions. Efficient administration lays the groundwork for innovation and continuous improvement [19], that is according to the study of Okokoyo [2024]. Garnering positive results to these literatures, it has come to shed a common understanding on how RPA could potentially benefit the process of organizational institutions. In another study, Gunawan et al. [2024] discussed that RPA helps the organization streamline the

business process by replacing manual effort that has low value-added, with a program that can increase effectiveness and efficiency [9].

Robotics automation can help reduce time spent on tasks by freeing up teachers to focus more on teaching and engaging students [16]. However, as elaborated by Madaev et al. [2023], it is also important to consider that the implementation of these technologies requires specialized skills, which means that investment in staff training may be necessary to make significant progress[16].

RPA has displayed interestingly positive feedback especially on the educational sector as it is a great factor to consider applying digitalization to current practices. The studies gave importance as to how the performance of automation included several explorations to managing the process in the institutions and in comprehending systematic method that can help utilize long-term educational and continuing outcomes. The mentioned literature suggests that the process of automating operational tasks that required hands-on employee skills, is significant when maximized and used with methodologies that provide lasting impact.

2.1.3 Frameworks/Methodology for RPA Implementation

Robotic Process Automation (RPA) is an essential tool in today's modern world because it reduces human errors and increases the accuracy and boosts the overall efficiency of the business operation. Implementing an RPA solution requires an organized structure of methodology or framework to allow organized and successful implementation.

Emerging software technology has since guided business operations to perform relative tasks that mimic human actions, but the main problem is the inability for organizations to adapt and identify processes that are suitable for RPA [4]. Various approaches were used in implementing RPA, which many reports were believed to have resulted as 'failed' [12]. In a literature review of Herm et al. [2023] the authors presented a three-phased framework, it includes Initialization, Implementation and Scaling. This framework was developed to provide enough guidelines and

offer flexibility to complex environments in an organization. The framework were found to maximize the benefits of automation producing structured management and support to its implementation.

Following a framework that aimed to relieve employees to tedious tasks, Bédard et al. [2024] introduces a new form of methodology called Rule-based robotic process analysis (RRPA), designed to help automation specialists to determine which process is good for automation. The RRPA methodology was evaluated in two criteria; the first one is the feasibility, which measures how effective the process to be automated, the second one is the relevance, to determine if automating the process provides a significant value[4]. The results gave a lasting impact as to how effective and efficient the software robots perform while lessening error-prone tasks.

Conversely, a comparative study was conducted to clarify insights that were gained to cover the practices and effectiveness of adapting the use of automation technology. The unison of Agile and DevOps methodology highlighted the revolutionary potential of RPA in improving testing scalability, accuracy, and efficiency [20]. Agile methodology supports an iterative approach that ensures continuous development and testing across the whole lifecycle of the project [11]. The authors Rami et al [2024], added that the development process using the methodology gains many advantages over traditional testing techniques and iterative development by shortening delivery times, and guaranteeing the release of high-caliber software offerings.

Additionally, the huge popularity of RPA has focused on combining RPA and BPM to maximize the potential of both approach [10]. The integration of RPA using BPM lifecycle were found to be essential as it ensures that all the automated processes are well-defined, efficient and aligned with the business goals. In the review study of Gupta et al. [2024], the combined approach allowed the researchers to evaluate automation in assessing the accuracy and efficiency of implemented RPA-driven solutions. A road map for future decisions were developed on each

identified issues as well as being able to provide modifications to business processes in the future automation perspectives.

In essence, while RPA technology poses a lot of advantages, some organization fails to adapt, due one major challenge in implementing an RPA: the failure to identify which task/process is automatable. Many studies developed a consolidated framework for the implementation of RPA projects using a design science research methodology while different framework consisting of different stages, provided structured yet flexible guidance to accommodate the complexities of diverse business environments. Despite challenges in task selection, evolving methodologies like RRPA Bédard et al., and the Three-Phased Approach of Herm et al., has proven to be effective in identifying, implementing, and scaling automation effectively. Many study review and literature of added that RPA technology, if accurately used, only succeeds when guided by structured frameworks like BPM, Agile, and DevOps methodologies. These methods not only help grow the business but provide a long-lasting impact on the efficiency of business operations.

2.1.4 Assessing Accuracy and Efficiency in the Implementation of RPA

Robotic Process Automation adopted in small and up to large scale enterprises, is beneficial when assessed in its efficiency and accuracy. The implemented automation solution ensures that these sectors achieve the reliable outcomes.

Apart from emphasizing the importance of human adaptation and acceptance to such computerized systems, the use of strategic management help develop a strong foundation to evaluate system performances like RPA. In an interview to identify the possibility of implementing RPA and its challenges to an organization, Luttinen [2022] used Critical Success Factors to identify the most important attributes and the detrimental ones in the application of RPA to an organization. Critical Success Factors (CSF) are specific elements or action areas a business, team, or department must focus on and successfully implement to reach its strategic objectives [26]. Although the organization's units had setbacks in the RPA implementation project because of believing it was

unsuccessful, Luttinen elaborated that RPA is something that the organization would benefit from in the future in a large scale [15]. The results expanded the perspective of the existing knowledge about RPA and also emphasizing its successful implementation in the organization.

Moreover, the potential of RPA does not only oversees in reshaping organizational forms and actions but also recalibrates productivity paradigms as discussed by Oyeniyi et al. [2024]. Their findings determined the significant efficiency gains and accuracy improvements. The researchers added that transformative potential of RPA spotlight the criticality of user acceptance and adaptation to these automated systems [8]. Most compellingly, perhaps, the research forecasted long-term implications of automation, envisioning a transformed profession most especially in accounting, characterized by strategic involvement and advisory expertise.

Accordingly, observing any potential bottlenecks or limitations in scalability, as well as assessing the impact on efficiency and accuracy on data extraction from invoices were done by Bhardwaj et al. [2024], the researchers designed a comprehensive analysis in 5 different scenarios related to Invoice-Processing tasks. The consistent performance of RPA underscores its reliability and precision, mitigating the risk of errors and ensuring high-quality outcomes [5]. The results not only displayed that RPA greatly improves accuracy and efficiency but also helps businesses save substantial productivity and time.

As discussed in the early literatures, RPA not only proved efficient results but rather optimizes workflow [18]. In fact, there are many forms of automation testing that uses Programming Languages and software automation systems as a tool. Some of these are Jacada, Automation Anywhere, Python and JavaScript that were found to perform more effectively [27]. To understand more about this concept, a study of Ogli [2024] uses Python scripts as a tool to automate and optimize workflow. The software robot were able to run seamlessly on multiple operating systems, ensuring broad applicability [18]. With this being said, some forms of

automation tools run dynamically like Python and were found to be successful as this provides the syntax to process workflow optimization.

Notably, two local studies from the Camarines Sur Polytechnic Colleges proved that use of RPA 1.0 using another automation tool like Microsoft Power Automate Desktop (PAD) were also essential. These studies produced an outcome where the results proved accurate and efficient use of RPA 1.0 using Microsoft Power Automate Desktop (PAD). Bonot et. Al [2024] reduced the processing time of enrollment procedures by 22.28 seconds in the tested set of cases [6], while, in another study conducted by Candelaria et al [2024], streamlining grade related procedures using RPA-PAD also achieved a success rate [7]. These results were achieved using the software robot that performed a positive approach as the integration RPA 1.0 significantly improved efficiency and accuracy, reducing task times in the academic institution.

The recent development and adaptation to automation technology reveals that many studies acknowledge various forms of automation tools and evaluation measures to understand better outcomes in every work environment. Such automation tools include heavy programming like Python and Javascript, and light user-friendly application like Microsoft Power Automate Desktop. Evaluation measures used from each study were deemed to recognize potential factors associated to the effectiveness and implementation of RPA. This indicates that there is increasing reliance on automation technology's ability to increase operational excellence in a variety of technological advancements in every workplace.

2.1.5 Impact of RPA 1.0 in generation of workload

In the modern pursuit of operational efficiency and automation, robotic process automation (RPA) emerges as a crucial technological development that enables businesses to use software robots to automate repetitive operations [28].

The study of Robotic Process Automation (RPA) technology is becoming increasingly relevant due to the rise in data and repetitive task that impact businesses' efficiency. According to Andrade et al. [2022], the best solution to handle processes with well-structured rules and repetitive tasks is by implementing RPA [1]. While it is given that these tools are primarily aimed at performing repetitive tasks, the efficiency of their work can be improved by using RPA tools to lower costs compared to other approaches. Accordingly, as discussed in the literatures of Okokoyo, these solutions have the potential to completely transform school administration [19]. Considering the challenges in every administrative operation, tasks which are both time-consuming and repetitive, may be made more efficient by using RPA solutions [25], as discussed in the study of Zahurin et al. [2024].

Educators are given a sense of worth and recognition when they are able to focus on teaching rather than performing administrative tasks [25]. They have proven to be the living embodiment of how much paper works relates to deadlines and problems, it is why a shift to RPA were suggested to ease work burnouts. Sharma et al. [2022] states that RPA significantly reduces workload by automating repetitive tasks, minimizing human intervention, and enhancing processing speed [22]. With dynamic use of software simulation or by using heavy programming languages as automation tools, not only eases current their set-up but bridges advantages for generating workloads.

In another study the researchers discuss that RPA extended beyond quantitative gains; it facilitated better utilization of human resources by freeing employees from repetitive tasks[17], Manfredotti [2023] pointed that using RPA will allow them to focus on a more strategic roles while demonstrating substantial increase in the volume of automation-generated results. Additionally, taking advantage of new opportunities and maintain market competitiveness could be gone beyond in fostering personal development as discussed by Trajkovska et al. [2024]. The impact of RPA could also improve organizational effectiveness, which helps to produce an advantage move

beyond operational efficiency; helps with the digital transformation[23]. Companies can gain deeper insights into their operations and remain competitive in a fast-paced operational landscape.

In conclusion, the studies and literature assessed by the researchers display holistic data that were reliable based on the relevant findings. When partnered with correct methodologies and frameworks RPA integration is an opportunity to overcome business and operational challenges. Studies have enough proof that the use of RPA is rooted to enhance life-long work-related operations that also improves flexibility to adapt to market changes. Furthermore, as automation technologies advance, organizations that successfully employ RPA also employs long-term growth and competitive advantage.

2.2 Synthesis of the State-of-the-Art

This synthesis included a thorough review and analysis of both local and foreign literatures that are found influential to the focus of the study. The literatures were known to have similarities and were highlighted in several key aspects.

The literatures achieved by the researchers encompasses similarities and differences in the context of this study. As one might know, workload schedules must be well organized in order to improve productivity as well as resource management. The generation of time-table provided a strong collective similarity and were discussed in studies of Vikulin et al. [2020] and Xue et al. [2024] [14, 24], where the literatures emphasize the huge involvement of time and schedules that requires precision. The importance of enhancing schedule generation and ensuring accuracy and efficiency in workload distribution is significant even in the present study, as well as understanding how it affects work performances. The time-table mentioned by previous studies discussed the importance of work-schedules, while the deployment of RPA on these intricate tasks were focused and centered on providing output-based result from the purpose of the study. As found in the study Jomuad et al. [2021], workload imbalance arguably can lead to faculty burnout, and emphasizing the importance of automated solutions that does not only enhance task distribution but helps reduce

stress among educators[13]. While these literatures emphasizes the effects of workload, the development that require keen understanding of applying automation measures in every step of plotting each schedule. These reasons are marked to differ from the application of present study which falls under the technicality of the problem in plotting these known work schedules.

In an educational landscape, technological adaptations has reshaped an academic operational tasks particularly in managing repetitive and time-consuming tasks. Studies from Zahurin et al. [2024], Selamat et al.[2022], Okokoyo [2024], and Gunawan et al [2023] emphasize that manual workload processes carry heavy burdens on employee that often results in stress or inefficiency. Additionally, the promising effect of RPA were highlighted as a solution similarly determined in the objectives of this study. However, the studies display differences such as in human-centered approach to adaptation in organizational needs and the fear of strategically involving automation-driven operations.

Meanwhile, different methodologies from each studies suggests a structured methodology to ensure successful integration of RPA. Studies from Herm et al. [2023] and Bédard et al. [2024], stresses systematic assessment by introducing new methodology to ensure that the automation will deliver meaningful value. Approaches of Agile, DevOps and BPM frameworks were also highlighted as it delivers an iterative development, continuous testing to enhance scalability and efficiency which similarly connect to the purpose of this study. While these methodologies offer the same goal of improving RPA, the methodologies differ based on their focus: some prioritize structured phases while other literatures emphasizes process analysis.

Empirical evidence demonstrating how RPA improves efficiency is frequently evaluated in terms of accuracy, efficiency, and possible challenges. Luttinen [2022] and Oyeniyi et al. [2024] highlight the importance of measuring automation outcomes while acknowledging implementation challenges [8, 15]. These challenges correlated to heavy programming used by Ogli [2024]. While Ogli used Python as an automation tool, Candelaria et al. [2024] and Bonot et al. [2024] used a

user-friendly approach like Microsoft PAD similar in this study's context. The differences were recognized based on the use of automation tools and implementation approaches that hinders operational effectiveness.

Lastly, the study's objective to improving efficiency and allowing employees to focus on higher-value responsibilities, were emphasized similarly to Andrade et al. [2022] and Manfredotti [2023]. According to them RPA is particularly effective for processes with well-structured rules. Zahurin et al. [2024] and Sharma et al. [2022] on the other hand highlighted its role in minimizing human intervention and speeding up processing. Collectively, these studies affirm that RPA optimizes workflows, improves resource utilization and mitigates the burden of repetitive administrative tasks similar to institutional environment. However, these studies differ in terms work setting when focused on reducing administrative burdens and producing long-term operational improvements.

2.3 Gap Bridged by the Study

Based on the findings found in the related studies and literatures, each study has shown a distinct approach in ways that it differs in the current study. These aspects were identified and highlighted as a gap in each area correlating to previous literatures.

Studies from Vikulin, M., et al., [2020] focused on the impact of time schedules to an educator's productivity while Jomuad et al. [2021] and Zahurin et al. [2024] emphasizes the negative effects and imbalances. These existing literatures extensively discusses the importance of well-structured schedules to an employee, where each result often adapts on a human-centered approach leaving a limited discussion on technical implementation of automation. In addition, the application of workload distribution systems provides a noticeable gap leaving a limited discussion on technical implementation of automation especially related to faculty workload-schedule plotting.

The role of RPA in the abovementioned literature suggests the potential of automation to alleviate stress and enhance efficiency using different methodologies both old and new. The authors-made frameworks, like Rule-based Robotic Process Analysis (RRPA) of Bédard et al. [2024], were mainly concerned with determining the value but lacks application of RPA. Furthermore, many automation tools were introduced such as using heavy-programming skills like Python but lacks discussion to underlying tasks that is tailored in an institutional setting.

Thus, this study aims to fill these gaps by developing an RPA system focused on structured-based approaches capable of applying automation in plotting faculty workloads. The use of BPM framework will be adapted in this study as it is believed to improve automation's overall effectiveness and further align them with the organizational goals of the College of Computer Studies. This reason bridges conceptual insight from past studies with a concrete application that addresses both the operational and technical generation of *automating the procedure of workload schedule* using Microsoft Power Automate Desktop.

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CHAPTER 3

TECHNICAL BACKGROUND

This chapter discusses the research methodology, research methods and statistical tools. Additionally, the use of Business Process Management (BPM) Framework and its description is discussed to accomplish the objectives of this study.

3.1 Business Process Management Framework

In today's fast-paced and highly competitive business environment, organizations are continuously looking for techniques to increase their efficiency, reduce expenses and provide more value to their customers. One of the most effective techniques is by implementing the Business Process Management (BPM) Framework.

The Business Process Management (BPM) Framework is a structured approach that guides the continuous improvement of business processes. It allows the organization to align their operations to the general business goals, ensuring that every process contributes to the strategic objectives. By using Business Process Management (BPM), organizations can continue to expand their view into their processes, identify areas of inefficiency and implement improvements that may result in improved performance, lower expenses, and increased customer satisfaction [3].

In this study, the BPM framework is adopted to effectively analyze, model and evaluate the process of workload schedule generation within the College of Computer Studies (CCS). This framework can provide the researchers with a clear and complete perspective of existing workflows, allowing them to identify operational bottlenecks and develop more efficient solutions. This approach enables the study to create a scalable and adaptive model that can support future improvements, such as integrating RPA 1.0 into workload schedule.

The BPM Framework is illustrated in Figure 1. It is based on the article of Dumas et al. [2012], which highlights the key stages of the whole BPM lifecycle [3].

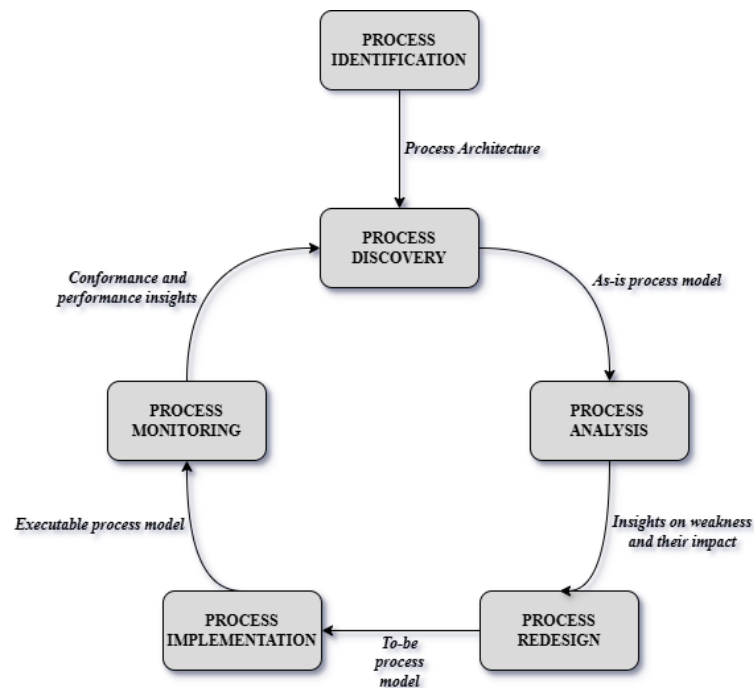


Figure 1. Business Process Management Lifecycle [3].

The following discussion provides a conceptual overview of each stage of the adopted BPM lifecycle and discusses how these phases were used in the study. Adapted phases are described as follows:

Process Identification. This phase involves identifying and defining the processes relevant to a business issue [3]. In the context of this study, the researchers identified the processes involved in the generation of workload schedules in the faculty. The generation of workload schedules is an important document that includes a specific format registered as an ISO of the college. It is a crucial document consisting of relevant information from the faculty which also can be time-consuming in the operational tasks to the college. To gain a deeper understanding of the existing processes involved in the making of workload schedules, the researchers used a semi-structured interview and observation to comprehend a solution in applying automation measures.

Process Discovery. This phase focuses in collecting data regarding the current (as-is model) processes and document them using process models [3]. In order to understand the current process

of workload plotting schedules, an As-Is model were mapped out in understanding the steps and tasks involved. Through observations and interviews, the process in creating one schedule helped the researchers identify possible errors and how concerns are faced in plotting each schedule. The As-Is Model helps to visualize and understand how each step happens.

Process Analysis. The process analysis goal is to identify and understand the problems like uncover issues, this allows organizations to prioritize issues based on their impact and urgency [3] . In this phase, the collected data in connection to plotting schedules were analyzed as to how the processes are being done in the college. Using the automatability assessment and scoring sheet, this helps the researchers determine tasks and prioritize areas needed to be improved all while identifying RPA solutions.

Process Redesign. This phase includes the redesign or the improved (to-be model) versions of the processes to solve identified issues. Redesign options are developed and compared to decide the best solution that meets performance goals [3] . Using the result from the analysis phase, the researchers mapped out a ‘To-Be’ process guide using BPMN Model aligned with the purpose of integrating RPA 1.0 using Microsoft Power Automate Desktop (PAD) to the tasks discovered.

Process Implementation. This phase includes managing changes to ensure a smooth transition from As-Is mode to the To-Be model [3]. In this study, the redesigned (To-Be) automated process model consisting of extracting data from Subject Load Notice (SLN), plotting each schedule of the faculty, and sending each generated workload to the instructor were implemented by following a standardized approach using Microsoft PAD.

Process Monitoring. Once the to-be model is implemented, the process is continuously observed and evaluated using evaluation metrics. Monitoring helps detect issues early and support ongoing improvement [3] . The evaluation of the automated process using predefined metrics is monitored to measure efficiency of using RPA and the accuracy of each generated result.

3.2 Research Design

Based on the goals discussed in the last chapter, the objectives presented a multi-method approach in achieving the desired results. Multimethod research is research that uses multiple forms of qualitative data or multiple forms of quantitative data [4]. Qualitative approaches is used for interviews, observation journal and related documents. These allowed the researchers to interpret contexts and phenomena aligning to the perspective of the study. While Quantitative approaches is used to indicators requiring numerical interpretation such as rating scale, Linkert scale, and score-based rubric, allowed the researchers, to measure and confirm statistical data to produce a clearer evaluation and analysis.

This research design is found to be essential in achieving the objectives of the study. The goal of Quali-quantitative research is not to replace either of these approaches but rather to draw from the strengths and minimize the weaknesses of both single research studies and across studies. The use of the combination approach improved the reliability of the findings by following the Business Process Management Framework.

3.3 Research Method

To achieve the goals of this study, this section will discuss the implementation of qualitative and quantitative approach, the use of instrument and data gathering procedures in achieving the chosen approach.

3.3.1 Data Sources

This section presents both the primary and secondary data sources that were identified by the researchers to meet the purpose of this study.

A. Faculty. These persons are the primary source of data consisting of first-hand knowledge about their timetable. They are directly the main subject to provide essential information as to how they

manually plot their own subject load and schedule. Through observations and interview, this helped the researchers identify specific solutions in applying automation measures.

B. Dean's Staff. These are also considered as the primary sources that provide essential insights on how procedural aspects in the context of workload schedule are done. The role of the Dean Staff allows the researchers to determine the managerial processes present in the college, including the importance of the faculty's subject load.

C. Subject Load Notice (SLN) Document. SLN Document serves as the secondary sources in this study that supports the very essence of plotting a faculty schedule. SLN provides accurate and reliable references for a faculty's schedule containing data such as the list of the faculty, summary of their subject load, designation, and the current academic year of each faculty member needed in the workload schedule.

D. Workload Schedule Template. The Workload Schedule Template is considered as secondary sources that contains information such as Time, Day, Room, Subject Code, Program Year and Section, Current Semester, Name of Faculty and the Dean. It is also an ISO registered document where the faculty plots their schedule manually.

These sources provided essential information in bringing clarity and legitimacy of data that were gathered by the researchers. Along with these sources, the faculty were identified as a critical entity to provide essential knowledge about the plotting process and creating automated solutions using Microsoft PAD.

Respondents and Sampling Technique

This section discusses the respondent and the sampling technique used by the researchers to achieve the goals of this study.

To ensure that the results reflect a relevant and credible source aligned in the context of applying automation to the manual practices, the faculty consisting of Regular,

Part-Timer, and Contract of Service were considered to as the main respondent. The respondents refer to the individual who participates in research by completing data collection tools [1]. These provide credibility as the faculty schedule varies from each designation. Meanwhile, to provide a fair representation of each faculty, the researchers used Stratified Sampling to ensure that each subgroup is adequately represented [5]. The stratified sampling formula is shown below.

$$p = \frac{N_p}{n} * 100$$

Where:

p = Percentage

N_p = subgroup population

n = total population

The use of stratified sampling formula helped gather the percentage of each group (Regular, Part-Timer, and Contract of Service) including the dean staff to ensure that each group is proportion to the total number of the population. Table 1 shows the sample size through the application of proportionate stratified random sampling.

Table 1: Target Sample Size

RESPONDENTS	ACTUAL POPULATION	PERCENTAGE	TARGET SAMPLE
Dean Staff	3	5	3
Regular Faculty	25	38	22
Contract of Service	23	35	20
Administrative Faculty (Part-Timer)	15	22	13
TOTAL	66	100	57

The sample size identified in the analysis of creating faculty workloads of the CCS is shown in Table 3. The actual population was identified from the School Year 2024-2025 with a total of 63 population. Using Slovin's Formula with a 5% margin of error, each subgroup achieves a total sample size of fifty seven (57). All dean staff, while twenty-two (22) respondents from

regular faculty, Twenty (20) from Contract of Service, and Thirteen (13) from Part-Timer of the population will be sampled. These numbers were examined to gather general insights of how effective and accurate the use of automation in the generation of workload.

3.3.2 Instrument

To effectively gather data for this study, a combination of research instruments were used. Each instrument has been combined with an appropriate measurement tool to ensure an accurate quantitative and qualitative data collection. Table 1 below shows the instruments and their corresponding measurement tools:

Table 2: Instrument and Measurement Tools

INSTRUMENT	MEASUREMENT TOOL
Interview Guide	Open-ended questions
Observation Journal	Rating scale, Linkert scale
Automatability Score Sheet	Score-based rubric, skill matrix

A. Interview Guide. Interviews are standardized or structured so that the same list of open- and close-ended questions is posed to each respondent [1]. In this study, the researchers were guided by semi-structure interviews to guarantee detailed information and thoroughness of the data being gathered. The interview helped the researchers analyze effective measures in understanding the documents involved in the plotting process of the faculty.

B. Observation Journal. Observation allows researchers to directly witness activities rather than relying solely on secondary accounts [6]. In this study, observation journal were used to provide accurate data such as the performance and issues tackled in each manual task. The observation helped the researchers gain first-hand insights on how the faculty and dean's staff interacts with software applications in doing document requirements and helps the researchers to note issues faced in the plotting process.

C. Simulation Journal. Simulation is a research or teaching technique that reproduces actual events and processes under test conditions [7]. In this study, the time-efficiency were measured by identifying the difference of both the manual practices and the simulated automated solution. The data gathered from the journal will help support the study by identifying the time-saved with integrating RPA solutions.

D. Inspection Journal. Inspection is a careful study or detailed scrutiny targeted towards critical understanding or decisions [8]. In this study, the simulated automated solutions were assessed to identify the accuracy of each document, specifically in plotting the day and time, entering the faculty and dean's name, the inputs of class program, year, section, and courses code. The inspection serves to provide accurate results whether the RPA generated the expected outcomes.

3.3.3 Data Gathering Procedure

In order to effectively recognize the practices being done in the college, the following procedures were done to collect the needed information in the context of this study.

A. Interview. The initial goal of conducting an interview was to comprehend the process of workload plotting of the faculty. This means that both the faculty and the dean's staff were interviewed according to the relevancy of the documents involved in the process. The use of semi-structured interviews helped the researchers follow a list of questions in relation to schedule plotting and freely discuss topics for improvement.

B. Observation. As the faculty demonstrates the process of making a workload schedule, observations were done by the researchers to recognize how each dependent task works. The observations help the researchers recognize how both the faculty and the software robot plots a schedule. Several patterns were also identified which helped the researchers to recognize human-computer interactions and capture solutions that can be added in the process.

C. Simulation. The automated solutions which were generated by the researchers are simulated to address the effectiveness of RPA. Using a pre-defined simulation journal, the researchers recorded

the time it takes for the software robot to plot the specified schedule of the faculty. This helped measure the time it takes to plot each schedule manually and the time saved by the software robot. Additionally, this also helped deploy test-simulation to document the behavior of software robots.

D. Inspection. After carefully identifying factors that affect the plotting of schedule processes, an inspection is done to gather additional information about the simulated automation. This means that the researchers record any conditions that may cause errors or inconsistencies. The inspection helps the researchers to ensure that the information of the faculty and the details mentioned are accurate in the proposed solutions.

E. Automatability Assessment. Automatability assessment-criteria is used to allow the researchers to analyze concepts and identify patterns involved in the integration of automation. The results helped identify automatable tasks and served as a guide to create the automated solution. The automatability assessment is discussed hereof:

Multi-Criteria Scoring Model

To offer a systematic approach in assessing the transformation of processes by integrating automation, the researchers used Multi-Scoring Model to analyze automatable tasks. Multi-Criteria Scoring Model is about choosing or ranking options when several criteria (or objectives) must be considered to rank and decide for alternatives [2]. Figure 2 represents the sample of scoring model utilized by the researchers. This was scored using the degree of scale shown in Table 2.

Table 3: Multi-Criteria Scoring Model

Task	Unstructure d	Standa r-dized	Repetitiv e	User Involvemen t	Voluminou s	Scor e	Degre e
Task 1	1	1	1	1	1	5	High
Task 2	1	1	0	1	1	4	High
Task 3..	0	0	1	1	1	3	Mediu m..

The figure above shows a sample for the criteria for assessing the evaluation of involved processes in the generation of workload plans of the faculty. These criteria were chosen based on

the factors that represent the relevancy of applying automation in tasks currently present in the college. Each criterion is determined as follows:

A. Unstructured Process. The workflow doesn't follow a consistent and orderly pattern, meaning the tasks is unclear, and lacks consistent execution of work.

B. Standardized. The workflow follows a uniform procedure which means that the task should be clearly defined, plotted, and has minimal issues.

C. Repetitive. The workflow is performed with the same actions, or the tasks are frequently done often leading to an increased frequency of errors.

D. User Involvement. The workflow is managed to a degree where human involvement is critically required for the task to be done.

E. Voluminous. The workflow shows a high volume of processing the task, which means that the task consumes a high volume of time and effort.

The table 3 will be used to add the identified tasks related to plotting each schedule. Using the digit 1 referred to as 'true', while 0 as 'false' [9], each criterion will be scored based on the tasks identified in the first column. and the total of each scored task is determined by adding the scores. After adding the digits, the degree is determined based on the range intervals identified in each task.

To classify where the average score falls within the measurement scale, the arithmetic mean was used as the primary basis for interpreting the scale values, as it represents the central tendency of the respondents' scores. The degree of automation of the manual workflow currently used in the faculty were evaluated based on The Range Scale for Degree of Automation. The scale helps provides clarity as to how the qualitative analysis done in the schedule generation were interpreted using scores[10]. The range scale for the degree of automation is shown in the Table 4.

Table 4: Range Scale for Degree of Automation

Scores	Range Intervals	Degree of Automation
5	3.66-5.0	High
4		
3	2.33-3.65	Medium
2	1.0-2.32	Low
1		

The table above shows the guidelines to score the multi-criteria scoring model, the range were obtained by subtracting the highest score of five (5) to the lowest score of one (1). The range were then divided on the number of degrees (High, Medium, and Low), to which were then equally placed. This classification of degrees allows a precise evaluation in the current manual input of schedules as the range intervals on the degree of automation were identified as 1.00-2.32 being the Lowest, 2.33-3.65 as Medium, and 3.66-5.00 as the Highest.

Value-Complexity Matrix for Process Prioritization

To prioritize which tasks to automate based on their value and difficulty, the Value-Complexity Matrix were used to provide analytical framework for decision-making. The Value-Complexity Matrix is a visual tool that helps to uncover tasks that deliver the most value in the very least effort [11]. In this study the term ‘complexity’ were used instead of ‘skill’ to reflect an accurate representation of complex tasks that are assessed for specific evaluation. The complexity refers to the ability of the defined task to be potentially automated, while the value refers to the importance of the task in the process.

The use of the Matrix helps to recognize tasks that can be redesigned in the process, allowing the researchers to identify and implement strategic efforts in applying RPA 1.0. Moreover, the value-complexity matrix will help the researchers justify the results of using value-complexity

matrix in assessing the tasks involved in the process of generating the workloads of the faculty.

The Value-Complexity Matrix is shown in Figure 2.

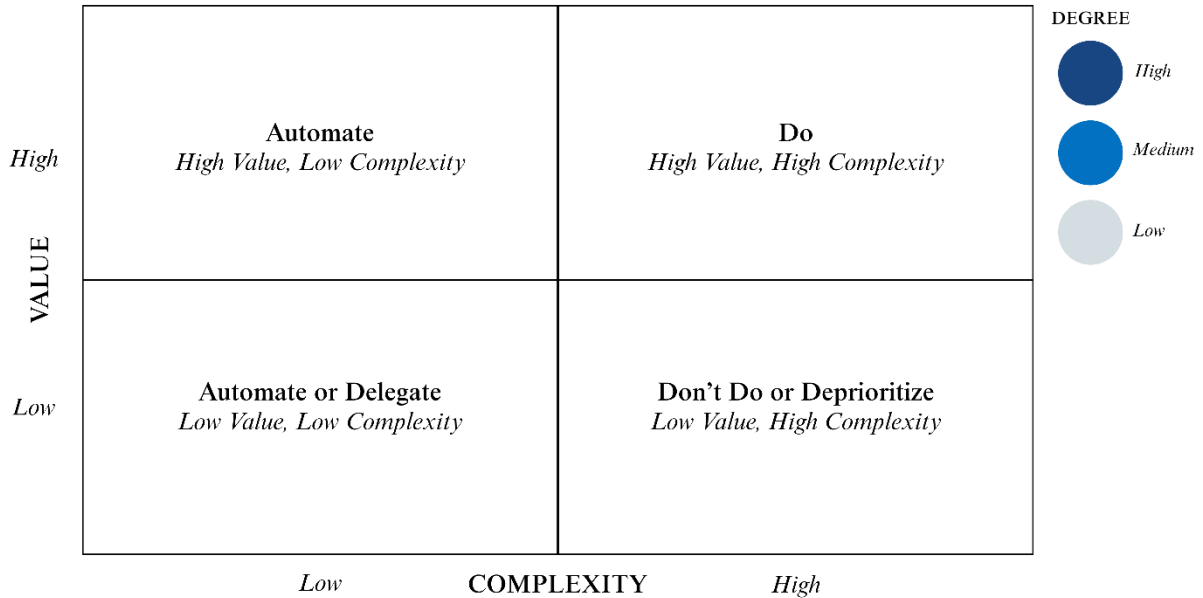


Figure 2. Value-Complexity Matrix for Process Prioritization

The figure above shows the table which will be used to plot each identified tasks based on its complexity and value. With four quadrants identified, the placement of the tasks will determine the automatability of tasks. The following quadrant are identified as:

A. High value, low complexity. These are tasks that indicates a top priority, as they demonstrate high perceived value and have relatively low skill which can be easy to automate.

B. High value, high complexity. These are tasks that offer high value but are more challenging to automate due to high complexity and system dependencies to human interaction.

C. Low value, low complexity. These are tasks that are easy to automate but provide limited value.

D. Low value, high complexity. These are tasks that are poor candidates for automation and are often deprioritized. They are difficult to implement as they impose a high complexity in a small value.

3.3.4 Data Analysis Procedure

This section discusses the interpretation of data, and systematic approach of the tools and the formulas used to support the objectives of applying Robotic Process Automation.

To examine and assess the automatability of the current process in generating the faculty workload schedule, the researchers used both quantitative and qualitative methods to recognize underlying factors in the plotting process. Using semi-structured interviews, the researchers identified that the process of generating the workload were based on a document generated by the Dean's staff, Subject Load Notice (SLN) document. Then using observation, the faculty were asked to plot each schedule. During this process, the researchers were able to graph the 'As-Is' process of schedule plotting using the Business Process Model Notation (BPMN).

After identifying the process, the researchers used the Multi-Criteria Scoring Model and Value-Complexity Matrix to assess the automatability of the current manual set up. These helped the researchers determine which tasks are subject to automation by scoring each identified task from the 'As-Is' model, recognizing its degree and identifying its priority. The gathered data helped initiate a 'To-Be' process model that serves as a guide in applying the automation solution developed using RPA. During this phase, the simulations and inspection of automation were journaled to help accumulate the time of manual plotting and the deployment of RPA.

To support progressive improvement of the current setting, the evaluation of the workload schedule is done to provide an accurate and fair representation of the gathered data. The evaluation measures in two sections: the Accuracy and the Efficiency. These metrics were evaluation based from the acquired data during the data gathering procedure. The formulae are presented hereof:

Evaluation Measures

To assess the effectiveness and performance of the software robot compared to the current practices, an evaluation were done to determine whether the automation delivers meaningful

improvements. In this study the performance of integrating RPA in the faculty workload generation, used a mathematical formula derived from *Arithmetic Mean*. Arithmetic Mean is used to treat all values equally, giving each data the same importance [12] This formula is used in two sections: accuracy and efficiency.

Accuracy

The percentage of the outcomes generated by the software robot, is used to measure the accuracy in terms of document correctness; time scheduling, faculty and dean's name, and the information gathered from the Subject Load Notice (SLN) document. To calculate the accuracy after integrating RPA on the faculty, the formula is presented as:

$$Accuracy = \frac{1}{n} \sum_{i=1}^n 1(\hat{y}_i - y_i) * 100$$

Where:

i = sample output

y_i = actual output with RPA

$\hat{y}_i$ = expected output

The Efficiency

After a thorough interview, the use of observation in data gathering procedure helps document the time for a faculty to plot their schedules. This helps the researchers to measure the time it takes for both faculty and the software robot to perform each task. To calculate the saved time for each process, the use of the following mathematical illustration is used to measure the efficiency of deploying automation to these defined tasks. The formula is presented as:

$$Time\ saved = \sum_{n=1}^n (t_i - s_i)$$

Where:

i -is the task from 1-n

t_i - time to complete the task without RPA

S_i - time to complete the task with RPA

(t_i-S_i) - calculates time saved for each task i

Arithmetic Mean

To calculate the average task of completion time, an expression $\sum x$ used to identify the sum of all values in the dataset, while n is the number of values in the dataset. This formula is used as the primary basis for interpreting the scale values, as it represents the central tendency of the respondents' scores. Furthermore, the following mathematical formula is used:

$$Average = \frac{\sum x}{n}$$

Notes

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CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the results and the analysis of using Business Process Management (BPM) Lifecycle framework. The phases of the BPM Lifecycle consisting of Process Identification, Process Discovery, Process Analysis, Process Testing, Process Implementation, and Process Monitoring were essential in supporting the study's objectives using Microsoft Power Automate Desktop (PAD).

4.1 Process Identification

The first component of the BPM Framework is finding processes important to the present operational flow. To gain insights on how the college generates their schedule manually, an interview guide and open-ended questions were used as a tool to effectively guarantee thoroughness of understanding the collected information.

4.1.1 Workload Generation

The current workload plan was recognized to show relevance in terms of reportorial requirements in the College of Computer Studies. In fact, like any operational tasks, the generation of a faculty workload schedule is essential as it does not only assign workload schedule but also aligns to resource utilization and time management of the faculty. Effective workload distribution not only helps stressed workers avoid burnout but also keeps them from feeling overworked in the first place [4].

The Business Process Model Notation (BPMN) was used to visually understand how the process of creating workload is done. Using the BPMN Model, the process was viewed in a structured approach identifying the interactions and decision points of the entities involved. The model was found to be essential to clearly see an overview of dependent tasks that exist, in what sequence the task falls into, and the potential to be automated realistically instead of having

assumptions. This also clarifies the roles of the individual and how the current structure of the process flows.

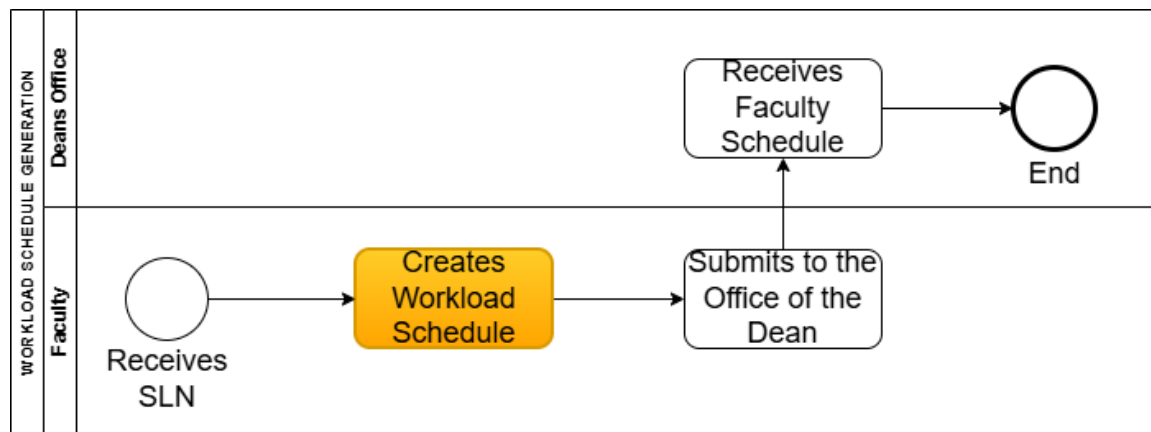


Figure 3. Process Overview of Schedule Generation Procedure

The process of generating a workload schedule is presented in Figure 3 in a series of actions intended to understand the current workflow of the college. This begins at receiving a Subject Load Notice (SLN), followed by creating the Workload Schedule by the faculty and then submitting the workload schedule to the Dean's Office.

Upon receiving the Subject Load for a semester (SLN), the faculty analyzes the document consisting of their names and their teaching loads. This triggers the input of creating the workload schedule subsequently provided in the SLN in accordance to the institutional guidelines. An interaction happens when the document is then to be submitted to the Office of the Deans for review and documentation. The logical flow of the process follows a series of sequential actions but has semi-structured approach when met specifically in creating a workload schedule (highlighted in yellow).

Research findings indicate that the current workload scheduling process is adequate in terms of structure but suffers from inefficiencies caused by manual operations. As highlighted in the study of Vikulin et al [2020], timetable scheduling is complex and complicated [2], the making of timetable requires a lot of time and effort especially in plotting each schedule accurately and

effectively. Manual operations such as these operational tasks may or may not inflict stress [3], that's why it is important to know the current effect to its employees.

This research suggests that there are strong opportunities for allowing practices that contribute to the improvement of the college. For a Faculty Member, an efficient automated system would lay benefits and ensure timely approved workload schedules. Additionally, automation could reduce repetition to manual related tasks as the adoption of an improved workload scheduling process could result to faculty satisfaction.

The analysis was limited to the College of Computer Studies, and the results may not fully represent the scheduling process of the other college with such factor that tackles only the generation of the workload schedules in the faculty of the college mentioned. The exploration of using automation were limited to the generation of Workload Schedule and not in other operational process currently done the college.

In summary, the current process revealed an inefficient manual workload scheduling process and is slowed down by human dependency and potential repetitive tasks. The BPMN diagram provides insights into workload scheduling workflow highlighting the benefits of adopting automation to enhance efficiency, accuracy and timelessness in creating the workload schedule.

4.2 Process Discovery

Following the observations and interviews done by the researchers, the operational processes were identified which show similarities and different approaches of the faculty such in generating their schedules. The process were mapped out to visually understand the process.

4.2.1 As-Is Workload Generation Process

An interview and observation were carried out in the College of Computer Studies department to gain an understanding of the existing procedure used in the development of workload schedules. Using Business Process Modeling Notation (BPMN), the tasks were visualized

illustrating the descriptive visualization of the Workload Schedule Generation of the faculty, the visualization of tasks is shown in Figure 4.

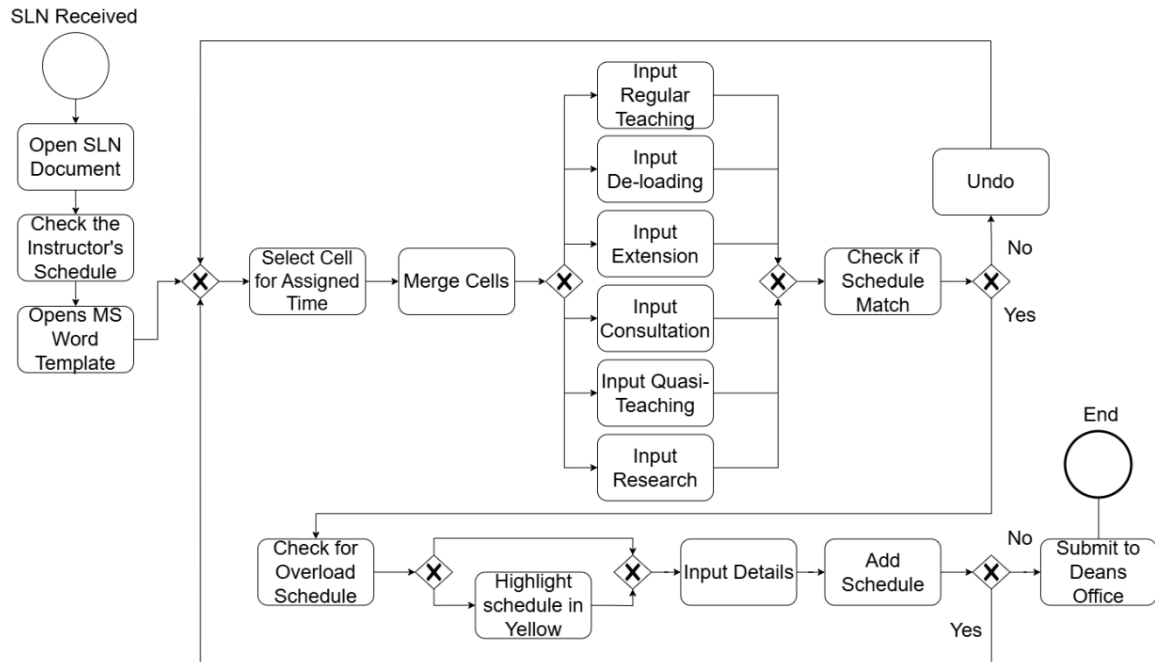


Figure 4. Descriptive As-Is Model Visualization of Workload Generation

These tasks revealed an approach that includes the primary input of the SLN Document in the process where the faculty interacts within applications to supply the necessary requirements needed. The flow of the process is discussed in letters:

A. Receive SLN. The process of schedule generation starts when the faculty receives the Subject Load Notice (SLN), a document that includes the faculty's subject load data. This serves as the official reference for the faculty's assigned subjects, teaching hours, workload schedules and other important data from a faculty.

B. Open SLN Document. This step happens when an instructor scans the document generated by the dean's staff. The document consists of all the faculty members and their academic year.

C. Checks instructors Schedule. In this process the instructors will check their names together with their teaching load, class times and subject assignment. This will be their basis on plotting their assigned schedule for the school year.

D. Open MS word document. The MS Word Document template is a document where the faculty plots. It is also an ISO standard of the college that contains pre-defined sections and formats of cell, header, tables, and the names. This ensures that each document exhibits uniformity and consistency in the department's document.

E. Select Cell for Assigned Time. This part of the process is where a faculty navigates through the template. During this process, the faculty will start to select the appropriate cell in the template where the information from the SLN file is to be entered.

F. Merge Cells. After selecting the cells for assigned time, the faculty merges the cells and adds the appropriate data from the SLN. The merging of cells marks the duration of the class where the initial time and end time are reflected.

G. Input Regular Teaching. Regular teachings is considered as the official teaching load of the faculty that contains the schedule assigned. These are identified leads included in the SLN document.

H. Input De-loading. De-loading is also considered as the official teaching load of the faculty that contains the schedule assigned. These are also identified loads included in the SLN document.

I. Input Extension. Extension is an additional schedule included in the workload schedule of the faculty. These schedules are added by the Regular Faculty and Contract of Service.

J. Input Consultation. Consultation is also an additional schedule included in the workload schedule of the faculty. These schedules are also added by Regular Faculty and Contract of Service.

K. Input Quasi-Teaching. Quasi-Teaching is also an additional schedule. Unlike other faculty, the Regular Faculty is the only one who adds the Quasi-Teaching schedule to the workload schedule.

L. Input Research. Research is also an additional schedule included in the workload schedule, added only by the Regular Faculty. These schedules are added by a faculty taking up post-graduate research schedules.

M. Check if schedule matches. After adding a schedule, a faculty member checks if the input matches the instructor's official subject load notice. This step measures if the plotted schedules have errors and ensure that the teaching loads are correctly reflected.

N. Undo. Since the process of plotting schedules are prone to human mistakes, the process of plotting schedule consists of undoing a task. Undo is done when the faculty needs to reverse a mistake without disrupting the template.

O. Check for Overload Schedule. A faculty that consists of more than 18 load units are faculty that plots overload schedules. These schedules are plotted only by a Regular working faculty that has de-loading or departmental designation.

P. Highlight Schedule with Yellow. When a regular faculty has overload teachings, they highlight it with yellow. Schedules highlighted with yellow are considered additional working hours.

Q. Input of Details. Once each schedule has been identified, the Subject Code, Program, Year, Section, and Room Details are entered in each merged cell. The details are set to spot time vacancy which will be helpful for adding other schedules.

R. Add Schedule. After adding the details of the overload assignments, the faculty then decides whether to plot another schedule depending on their schedule assignment.

S. Submit to the office of the deans. The completed workload schedule is then submitted to the Dean's Office for validation and approval. This finalizes the workload schedule.

Based on the discovery, the Subject Load Notice (SLN) plays a major role in Workload Schedule Generation process. Using As-Is model for the visualization of these tasks, it revealed that the steps in making Workload Schedule, shows an unstructured repetition which may reflect a huge amount of processing time. The researchers also discovered that while the study aims to navigate the manual process of generating workload schedules by applying automation for lesser human intervention, the integration of RPA will still require human actions such as adding schedules not indicated in the SLN file.

4.3 Process Analysis

After closely identifying and learning about the current workflow processes at the college, the researchers were able to identify the tasks that are subject to automation. Using Thematic Analysis for structured interpretation of patterns [1], the researchers extracted significant tasks that can be measured in the assessment. The analysis also helped the researchers uncover and measure the automatability of the tasks mapped in the As-Is model using Multi-Criteria Scoring Model and Value-Complexity Matrix, the assessment of each tasks were discussed in this section.

4.3.1 Automatability Assessment using Multi-Criteria Scoring Model

The process that includes the tasks identified were assessed using the Multi-Criteria Scoring Model, based on the As-Is Model shown in the Process Discovery Phase. The tasks column reveal an input of activity that were presented in Table 5 consisting of rows that are evaluated for clearer understanding. The interpretation of flow were presented in letters:

Table 5: Multi-Criteria Score Model for Automation

Task	Activity	Unstruc- tured	Standar- dized	Repe- titive	User Involve- ment	Volu- minou- s	Scor- e	Degre- e
A	Receive SLN	1	1	0	1	0	3	Mediu- m
B	Open SLN file	1	1	1	1	1	5	High
C	Checks Schedule	1	1	1	1	0	4	High
D	Open Template	1	1	1	1	0	4	High
E	Select Assigned Time	1	1	1	1	1	5	High
F	Merge Cells	1	1	1	1	1	5	High
G	<i>Input Regular Teaching</i>	1	1	1	1	1	5	High
	<i>Input De-loading</i>	1	1	1	1	1	5	High
	<i>Input Extension</i>	1	1	1	1	1	5	High
	<i>Input Consultation</i>	1	1	1	1	1	5	High
	<i>Input Quasi- Teaching</i>	1	1	1	1	1	5	High
	<i>Input Research</i>	1	1	1	1	1	5	High

Multi-Criteria Score Model for Automation

Task	Activity	Unstructured	Standardized	Repetitive	User Involvement	Voluntariness	Score	Degree
H	Check if Match	1	1	1	1	1	5	High
I	Undo	1	1	1	1	1	5	High
J	Identify Overload Schedule	1	1	1	1	1	5	High
K	.Highlight Overload in Yellow	1	1	1	1	1	5	High
L	Input Details	1	1	1	1	1	5	High
M	Add Schedule	1	1	1	1	1	5	High
N	Submit to Dean's Office	0	1	0	1	0	3	Medium

A. Receiving of SLN Document. As the initial step that triggers the generation of workload, the task is Scored as three (3), and has a degree to which is defined as medium. The Subject Load Notice (SLN) is an important document summarizing a faculty's workload schedule. This is created by the Program Chair and was submitted to the Office of the Dean for approval. The SLN Document, once received by the faculty, consists of the Subjects and Overload Teaching the instructor may or may have.

B. Open SLN File. The opening of SLN File requires human engagement where the file is needed to be clicked to check the issued schedule, which can highly suggest being automated. The score of five (5) with 'High' degree indicates that this task can be automated.

C. Checks Schedule. Checking schedules is a highly repetitive task which also requires human interaction by switching between software applications involved in plotting the schedules. The task is scored into five (5), the degree suggests that the checking of schedule is highly automatable.

D. Open Template. Opening of Template also requires human intervention as this requires clicking of the document to plot the schedules issued in the SLN file. This task is scored as five (5) and displays a 'High' degree of automatability.

E. Selection of Assigned Time. In this step the instructor is set to plot the schedule provided from the SLN. Scored 5 with a degree that corresponds to ‘High’ shows that the process is subject to automation as selecting assigned time is voluminous and may cause human error. The selection of assigned time is one crucial action which plots an instructor working schedule this requires a keen eye of detail to effectively merge the correct assigned schedule.

F. Merging of Cells. The merging of cells is done when a faculty has selected the assigned time from SLN. With a score of five (5) the degree revealed a ‘High’ degree of automation the task meets the criteria to be automated. Additionally, this task is one of the core repetitive task that are usually done by the faculty.

G. Input of Schedules. The schedules which include *Regular Teaching, De-loading, Extension, Consultation, Quasi-Teaching, Research*, are schedules that contributes to time-consumption shows a ‘High’ degree in the automatability assessment. These schedules meet the criteria for automation having a score of five (5) as well as having the most.

H. Checking if Schedule Matches. This part of the step is significant as it is an action that are usually done repetitively especially in plotting the instructor schedule. This step requires switching from one document to another to ensure that each schedule matches from the SLN. Scored as five (5), this step is interpreted to show a ‘High’ degree of automation and is subject to be automatable.

I. Undo. The ‘Undo’ task is an action that is unstructured since it varies among the faculty to be done. The task revealed a ‘High’ degree of automation with a score of five (5) which indicates that the task can be automated. Based on the analysis done by the researchers, this task are often done and repetitive when a mistake in the plotting is met.

J. Identifying Overload Schedule. Schedules in the college consist of overload that marks whether an instructor is a regular working faculty or not. This highlighted schedule in the document indicates that the faculty has an additional working hour which differs from other employee. Based on the analysis done by the researchers, the step is voluminous as it relates to giving a massive time

and effort in plotting a schedule. A score of five (5) indicates that this stage has a 'High' degree of automation.

K. Highlight Overload Schedule to Yellow. The task of highlighting schedule to yellow is a complex task where it requires a keen eye for the user to identify an overload schedule reveals an approach that shows that each criterion were marked scoring a total of five (5). The task reveals a 'High' automatability degree as it requires keen details of user involvement.

L. Input of Details. The schedule type, course code, block (section), and room are details that are inputted in the schedule. With a score of five (5), the activity demonstrates a high level of automatability as it indicates that labeling merged cells and filling with the necessary data takes a significant amount of time and work.

M. Add Schedule. Additional Schedules are also one of the tasks that recur and time-consuming and is usually done by the faculty, with a score of five (5), the process indicates a 'High' degree of automatability in the process selection.

N. Submitting of Documents to the Office of the Dean. After carefully plotting the schedules of the faculty, and after the details are satisfied, the Generated Workload Schedule are expected to be passed on to the Office of the Dean. This action is reflected to as a Medium Degree scored with three (3) as it requires human intervention before submitting to the Office of the Dean.

In summary, the findings reveal that the tasks associated with schedule generation are dependent on which each task relies heavily on the SLN. The SLN triggers the whole process where each task works after the other. While it is a requirement in the college to generate such documents, these task that are passed down to the faculty receives a lot of complains. The faculty also revealed that most of the tasks significantly show a huge amount of highly repetitive plotting which are time-consuming.

Using Multi-Criteria Scoring Model the scored tasks reveal degrees to which suggest that these tasks that show a 'High' degree can be automated using PAD. Moreover, the score of task

serves as a guide that helps the researchers in creating an alternative way of generating schedules by integrating RPA using Microsoft PAD.

4.3.2 Automatability Assessment using Value-Complexity Matrix

To effectively evaluate and prioritize the identified tasks based on their potential value and level of complexity, the value–complexity matrix was utilized to offer a visual input of which tasks are most suited for automation. Using the degree presented in the value-complexity matrix, process of Workload Generation were evaluated based on its value and complexity. The input of tasks in the matrix were plotted in a color-coded circle to determine the quadrant they belong to. The dark blue represents a high level of automatability, the medium blue represents medium level, while the light blue represents the low level of automatability. The assessment of the value-complexity matrix is shown in Figure 5 whereas the identified tasks were interpreted in sequence of the letters A, B, C and so on.

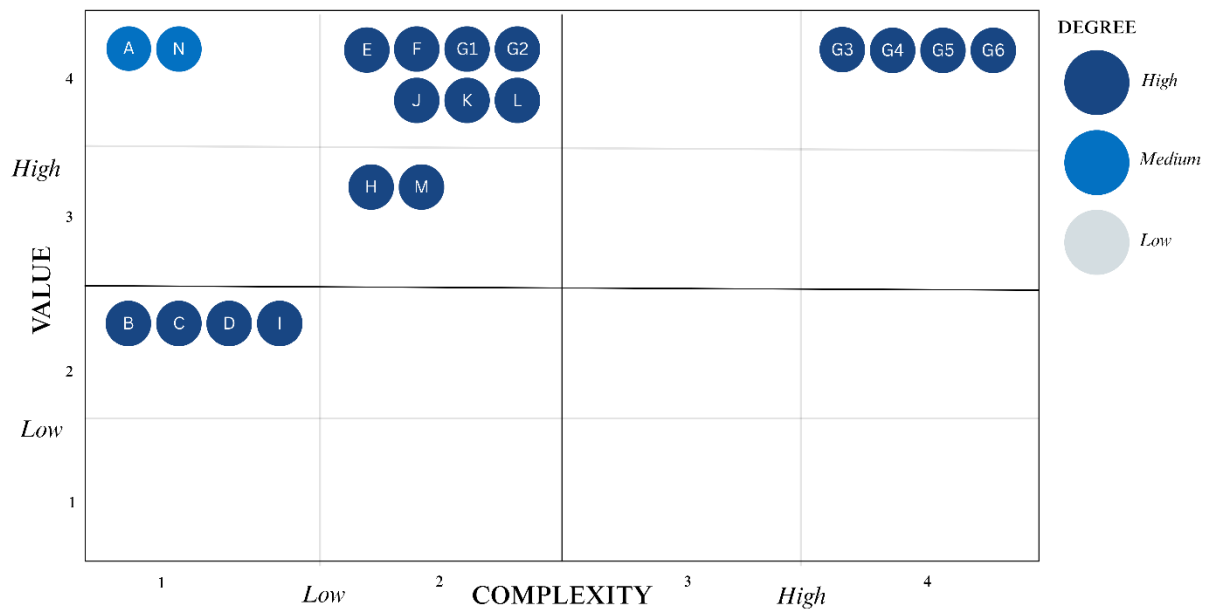


Figure 5. Value-Complexity Matrix for Process Prioritization

As shown in Figure 5, the tasks A, and N along with Tasks E, F, G1, G2, H, J, K, L, M are plotted in the 1st quadrant which shows that these tasks offer high value and is crucial in the plotting

process. These tasks indicate a top priority and can easily be automated due to their low complexity level. Tasks B, C, D, I, also display a low complexity level which can also be automated but these tasks offer a lesser value. Meanwhile in the far-right quadrant, G3, G4, G5, and G6, are tasks that have the most value than any other tasks. Although these tasks indicate a high value in the plotting process, these tasks also offer higher complexity in the process which require complex variables that are difficult for automated systems to comprehend. These tasks cannot be automate since it will require higher version of RPA.

Based from the results, the analysis of the tasks involved in the creation of workload schedule from each instructor consists of regular teaching loads while the permanent employee has de-loading designation. While it is possible to digitalize the process of generating workload schedules, the researchers identified that the SLN file containing the Faculty's Summary of Subject does not include: Research, Extension, Quasi-Teaching, and does not emphasize Overload teaching. Additionally, it is mentioned on previous chapters that the software robot from RPA 1.0 relies on a set of instructions and lacks decision-making capabilities. Hence, only the indicated schedules: Regular and unshaded Overload schedule will be plotted.

Moreover, the Value-Complexity Matrix helps support these findings by displaying the complexity of additional schedules in the process. This means that the tasks potentially involve human processing and the interaction of the software robot is limited in doing so. Additionally, it is discussed in the previous chapters that the use of automation, specifically RPA 1.0 is only limited on performing defined tasks which are set in the automation workflow. The process analysis helps identify these factors that were personally complained by the faculty to be inefficient.

4.4 Process Redesign

This phase is the continuation of the Discovery Phase that highlights the optimization and redesign of the previously visualized As-Is Model.

Based on the insights gathered from the process analysis, the redesigned process were modelled to guarantee that the procedure follow a logical and optimized process, which assures that the automation will produce significant structured outcomes and values that will continue to be beneficial in its performance. The redesigned process were mapped to visually tell the transition of integrating RPA 1.0 in the faculty manual-plotting set-up. The ‘To-Be’ process model ensures that these complex tasks are performed consistently with the help of the deploying software robot.

4.4.1 To-Be Model Workload Generation

With such factors discussed in the Analysis Phase, the researchers were able to identify and map a guide through on how the ‘To-Be’ process model will be generated. The process of tasks that were scored based on the automatability assessment and were mapped out in Figure 6 indicating the tasks that are to-be automated. The To-Be process model presented below is the suggested process using Microsoft PAD as an input to the process.

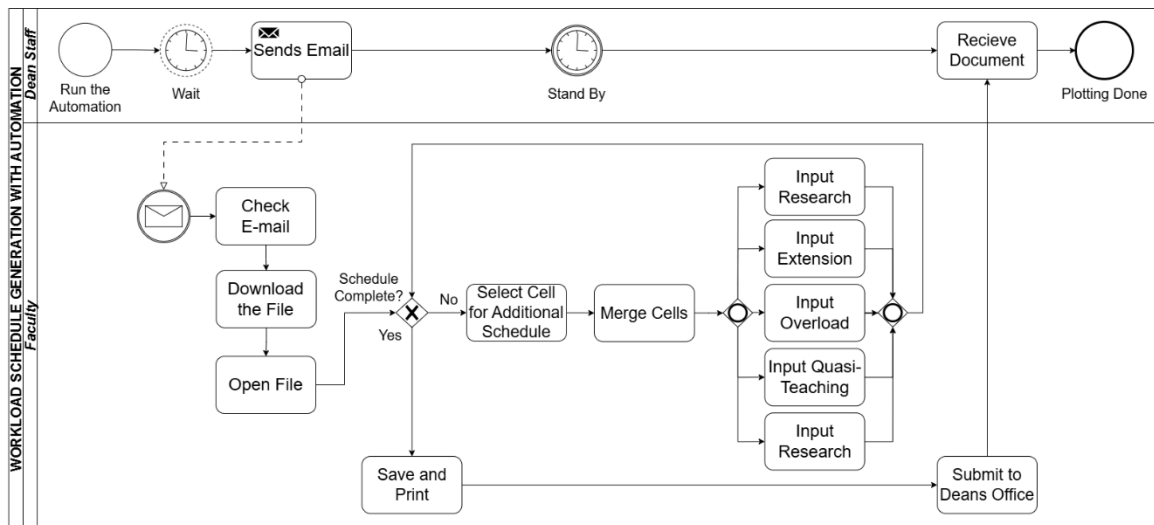


Figure 6. To-Be Process Model of Workload Generation

Figure 6 shows the To-Be process that aims to simplify the current process using automation. It was revealed in the Process Discovery phase that although the SLN document is the primary upon which an employee receives notice, the specific interaction between the process of

generating an instructor's workload follows an unstructured way and varies on how one creates their schedule. Although the general purpose of this study is to re-digitalize the current set-up of the faculty, the proposed process of workload generation will still require human intervention for the additional schedules not included in the SLN Summary. The logical flow of the To-Be process model were mapped out to effectively understand the structure and is discussed as follows.

A. Run the Automation. To perform the automation, the dean staff will have to run the automation by pressing the icon set in the desktop. This will trigger the automation to plot the regular workload of each instructor in a readily made template. After the authorized personnel run the application, the software robot plots each schedule of the faculty indicated in the SLN Summary which will be stored in a folder.

B. Sends E-mail. After the automation is launch, and after meeting all conditions, the software robot will send the plotted schedule that were saved as a worksheet. The document will be sent to each instructor for double checking of the document while the Dean's staff waits for the return of the plotted schedule in the Dean's office.

C. Checks E-mail. The faculty or instructor are the recipients which will receive the file containing the worksheet generated by the automation. This is the near-complete phase, during which regular working professors or instructors will add additional schedules, while Part-Timers and Contract of Service do not.

D. Download the File. Since this file lacks the additional schedule of the regular employee, the file will then be downloaded containing the automated-plotted schedules for their current year. The file will then be used to plot additional schedules that weren't indicated in the extracted data in the SLN file.

E. Open File. The faculty will open the file for additional plotting of schedules. If the faculty is a regular employee, manual plotting of *Research, Extension, Quasi-Teaching, and highlight of color*

for Overload will be required, while other employees like Part-Timer and Contract of Service will not.

F. Select Cell for Additional Schedule. After a faculty acquires the generated file of schedule, a selection of cells will be done to select schedules that weren't reflected in the worksheet.

G. Merge Cells. The merging of cells happens after selecting a vacant time. The merged cells marked the duration of the schedule.

H. Input of Additional Schedule. After selecting a cell in the worksheet, the input of additional schedules will be done by the faculty. These are schedules that weren't included in the SLN Summary File.

I. Save and Print. After plotting each additional schedule, and assessing if the schedule matches, the faculty will save and print the document for filling of other required information such as the wet signatories.

J. Submit to Dean's Office. The printed schedule will then be returned by the faculty to the dean's office for signatory of the workload schedule document.

K. Receive Workload Schedule Document. The document generated by the automation will be sent to the Office of the Dean's which will be signed for confirmatory of documents. This also marks the end of the process.

In summary, the initial step of clicking the icon shortcut will be used to start the automation, this action will launch the Virtual Machine used to directly trigger the automation process of generating each instructor's workload. After the software robot meets all conditions, each document will be sent directly to each instructor's email inboxes via email. Each identified tasks were converted and digitalized, as a refined process-solution that may help the current set up of the faculty. The researchers recognized the To-Be Process Model to display a structured handling of the software-generated document allowing to provide a more efficient workforce to both faculty and the dean's staff.

4.4.2 Technical (To-Be Process) Model of Workload Generation

To understand how the software robot performs each assignment, an accurate model showing the technical version of To-Be process model is used to present a much-detailed process of automation. The illustration shown in Figure 6 acts as a blueprint that translates into precise action of automation solutions and how the software robot performs each task based from the actions met.

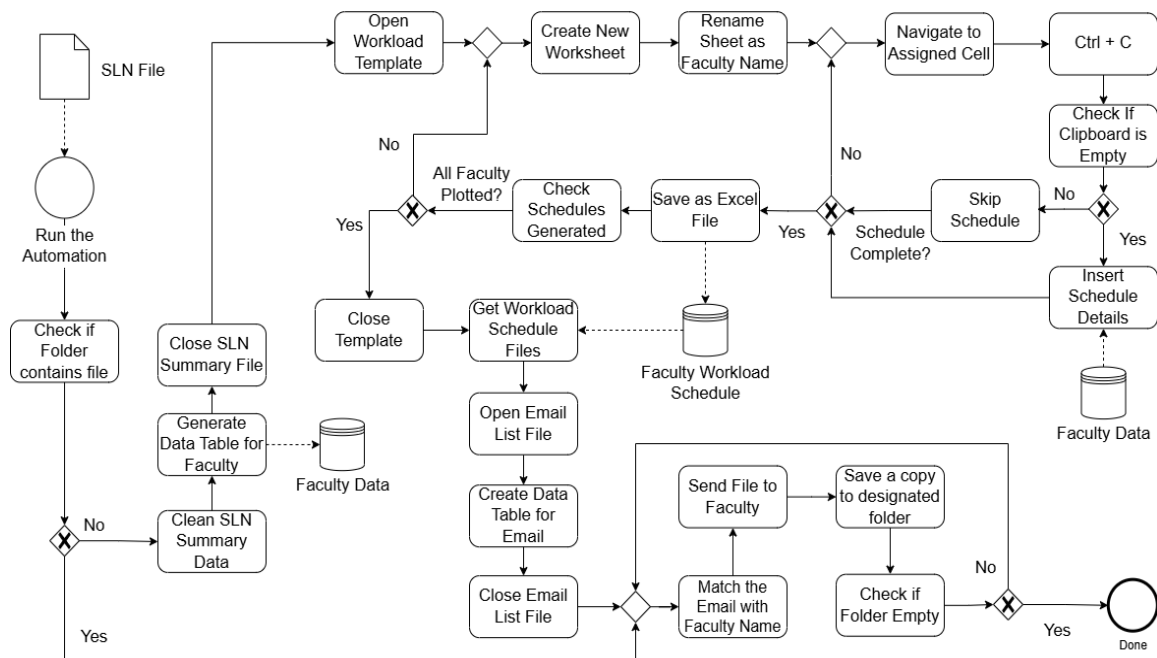


Figure 7. Technical Automation To-Be Process Model

The Technical Automation To-Be Process Model illustrated in Figure 7 shows the input of software robot in the process. The behavior reveals interactive-step-by-step process of the automation starting from preparing and cleaning the SLN summary data to sending the generated workload schedule to the faculty institutional email. The flow logic of tasks are discussed in letters.

A. Presence of SLN File. The To-Be process model will still require the presence of the SLN summary to extract the needed data in the plotting process. Since the SLN file contains all the

relevant data and has been the reference of most of the faculty, the file will still be the trigger of the whole process.

B. Run Automation/Click Desktop Icon. Upon the generation of SLN file, the dean staff will run the automation on-which the staff will have to wait for the virtual machine to launch. Then, the Microsoft PAD will automatically run to check whether the folder contains a file.

C. Clean SLN Summary. Since the SLN contains other components not required in the process, the software robot will store a duplicate data consisting of the needed information in the faculty schedule. This step ensures that the software robot will store only the relevant data needed in the plotting of schedules.

D. Generate Data Table. In this step, the software robot will store the extracted data from the SLN, containing the list of faculty members and their teaching loads. This helps the software robot perform a structured and organized plotting to easily populate necessary information in the workload schedule.

E. Close SLN File. After acquiring the data, the software robot will close the SLN file to prevent potential errors and conflicts. The closing of the file also helps reduce the consumption of the laptop's RAM as well as to prevent errors from performing in the users interface. This action also marks the next step to perform.

F. Open Workload Template. The same as how a faculty perform the process in the manual procedure, the software robot will also open a template where each faculty's schedule will be placed. The template is based on the original file but were saved in an Excel file consisting of pre-defined format: the file code, names, the hours and day (to name a few).

G. Create Worksheet. After opening the template, the software robot will create a new worksheet duplicate the template shown upon opening the file. The duplication of the template guarantees that the template will not be ruined, making sure that the template is ready for faculty schedule plotting.

H. Rename as Faculty Name. To easily identify which sheet will be plotted, the worksheet is renamed to as the name of the faculty. Renaming the sheet also helps recognize that the sheet in the template is generated only for the faculty.

I. Navigate to Assigned Cell. This step marks the start of the most repetitive task in the process; select the assigned cells where the schedule is to be inserted. Using a set of actions deployed by the researchers, the software robot will navigate to a cell based on the number of time that were extracted in the SLN file.

J. CTRL + C. Although this process doesn't look like much, this process has an important role for the software robot to perform. After selecting the cell, the CTRL + C action is done to check if that selected cells contain text. This text will define whether a schedule has been plotted or not.

K. Check if Clipboard is Empty. Once the CTRL + C action is performed, a clipboard is generated. The clipboard may contain a text that signifies if a schedule has been plotted.

L. Skip Schedule. The 'skip schedule' is an action set by the researchers after checking if the clipboard (Step K) contains a text. This is an error-handling carried out to prevent potential overlapping of schedules.

M. Insert Schedule Details. An input of a schedule will be done by the software robot, once the cell is identified to be empty. These details were written from the data table extracted from the SLN summary file.

N. Schedule Complete. Since the task of navigating and inserting details is repetitive, the automation will check if all the schedules from the Data table are plotted. If not, the software robot will repeat the action of navigating to a cell.

O. Save File. After checking if all the schedule has been plotted, the software robot will save the template as the faculty name. The file is stored in a designated location with all other schedules generated by the software robot.

P. Check if all Faculty Schedule is Generated. The data table that were generated by the software robot consisting of all the faculty data is checked to know if all the schedules are generated. If a faculty schedule has not been plotted, the software robot will create another worksheet (Step G).

Q. Close Template. After generating all the faculty schedule of the faculty, the software robot will close the template indicating that all the schedules were plotted. This action is done to continue the next step of the software robot.

R. Get Workload Schedule Files. The software robot will then retrieve all the files (worksheets) that were generated by the bot. Each file location were gathered to easily perform the next step.

S. Open E-mail List File. The software robot will open the e-mail list file containing all the email addresses of the faculty. The list will help the software robot send the generated schedules to each faculty inbox.

T. Create Data-Table for Emails. This action were set to extract the data and organize the list in a data table. These extracted emails are stored and will be used to populate the text required for sending the emails.

U. Close E-mail List File. After extracting the email data from the file, the software robot now has the needed data for sending emails. The automation will then close the file to perform the next step.

V. Match the Email with the Faculty Name. Having both Schedule and Faculty Email tables, the software robot now proceeds to match each schedule file (named as the faculty), to the email list. This action ensure that the schedule will be sent to the correct recipient.

W. Send File to the Faculty. After matching the data tables, the software robot will send the file using STMP action to easily send the generated schedule to the faculty. The faculty will then receive this file for reviewing.

X. Save a copy in a folder. The software robot will save a copy of the folder so that both the dean staff and the faculty have the same plotted schedule. The software robot will perform this action as a mark that the file has been sent to the faculty.

Y. Check if Folder is Empty. This action is set to identify whether the folder containing the schedule of the faculty is empty. If not, the automation will repeat to match the email and the file name (Step *W*) until the folder is empty.

Z. Done. A pop-up dialogue will appear notifying the dean staff that the automation is done. This marks the end of the automation process where-in the faculty is expected to receive the schedule.

The To-Be process using RPA suggests that these tasks may improve the process of creating and effectively disseminating faculty workload schedules. In summary, BPMN helped the researchers to map out and provide an overview of the process of generating the faculty workload schedule. To understand how the RPA will perform when deployed in a virtual machine, the Technical To-Be Process were mapped out to reveal the behavior of the software robot using a set of actions deployed by the researchers. In addition, the To-Be process is expected to reduce faculty-dean staff interaction as the process suggest being done within the dean's staff, allowing the faculty to make room for other relevant task. The researchers also believes that by following the implementation of RPA in process upgrades, the current practices in the college will offer an efficient way of handling workload schedule and in continuing a much faster result by simply applying automation to these technical steps.

4.5 Process Implementation

Following a thorough identification, analysis, and redesign of the current process discussed in the previous phases, the intended improvements are carried out to actual operations using the required hardware and software tools.

4.5.1 Hardware and Software Tool Specifications

The integration of RPA serves as a particular function in making certain that the automation is not only successful but is in accordance with the objectives and workload procedures[4]. These procedures were made possible by utilizing correct specifications to hardware tools in performing

required standards in the making of each workload schedule. It is mentioned in Chapter 2 that this study will focus on integrating automation using Microsoft PAD in the process but to implement the software robot ideally, a required hardware tools will be needed. The researchers identified several specifications needed to guarantee that the software and hardware tools will operate well. The input of these tools serves as a physical foundation necessary to interact between software applications and automation commands. The hardware specifications is presented in Table 6.

Table 6: Hardware Tools

Hardware Tools	Specifications
Laptop/Computer	16 GB RAM (8GB minimum)
	Quad-core Processor (equivalent to i3, AMD or higher)
	256 GB Free Storage
	1920 x 1080 Display Reslution
Internet Connection	Minimum of 10 mbps

Based on the article provided by Microsoft, the specifications needed to run Microsoft PAD will require a minimum of than 1.00 Ghz Processor, 1 GB of Random Access Memory (RAM), and 1 GB of free storage [5]. With the integration of RPA, the researchers recommend the specifications based on the identified requirements from VirtualBox [6], as well as considering the needed requirements to run multiple applications inside the virtual machine. The needed tools require: 16 GB RAM (or at least a minimum of 8 GB RAM), Quad-core processor (i3/AMD/ or higher), 256 free storage, 1920 x1080 display resolution and a working stable internet connection of at least 10 mbps.

These specifications for a desktop provided balance in terms of deploying activities and carrying out data to several workload generation processes especially when paired with heavy applications such as virtual machines. A working stable internet connection should also be considered to ensure that the PAD will launch and access other tools such as sending emails using the SMTP features. The hardware tools mentioned in Table 6 helps to contribute to a fully

functional integration of RPA in the workload schedule generation process. These specifications ensure that the required software tools will operate ideally. Moreover, the software specifications needed were presented in Table 7.

Table 7: Software Tools

Software Tools	Specifications
Operating System	Windows 10/Windows 11
Software Applications	VirtualBox
	Microsoft PAD
	Microsoft Excel
	Microsoft Word

To create and run specific flows, the use of software applications were used in deploying automation in the study. While the hardware tools were discussed as an essential foundation, the specific use of software tools also contributes importance such in processing the desired output of the study.

Based on the article provided by the Microsoft Company, an operating system running in Windows 10 or Windows 11 update, is needed to ensure the stability and compatibility of other software tools needed to run RPA. Moreover, the researchers identified four (4) software applications that will be needed: VirtualBox, Microsoft PAD, Microsoft Excel, Microsoft Word. These applications were necessary to perform actions set in the Microsoft PAD. Paired with the best hardware tools, the researchers believe that performing RPA in the process will provide a flexible and faster uptime work even in heavy computation of tasks necessary to run automation flows in the generation of workload schedules.

Table 7 shows the software specifications that were used as a tool to consistently deploy a fully running operating system specifically tethered in Windows 10 or the latest Windows 11 update. Using the the latest operating system, both hardware and software tool communicates well with software applications that allowed the researchers to execute each action in the necessary environment without delays. The use of Microsoft PAD as a core automation tool allowed the

researcher to run commands which reduces the complexity in the generation of each workload schedule. Lastly the use of Microsoft Excel, serves as a tool for processing both data extraction and plotting of faculty schedules. The use of these software applications assisted in keeping uniform records and tasks tailored in storing each data.

4.5.2 Integration of Automation using Microsoft PAD

Using the expected software and hardware specifications, the automation behaves seamlessly between software applications ensuring each schedule were inputted in seconds. The hardware tools helped carry the performance of the software robot to work and perform the desired results effectively as paired to each software applications. Additionally, the Virtual Machine helped perform the process without interrupting the main computer. The interaction of generating workload schedules were discussed in figures.

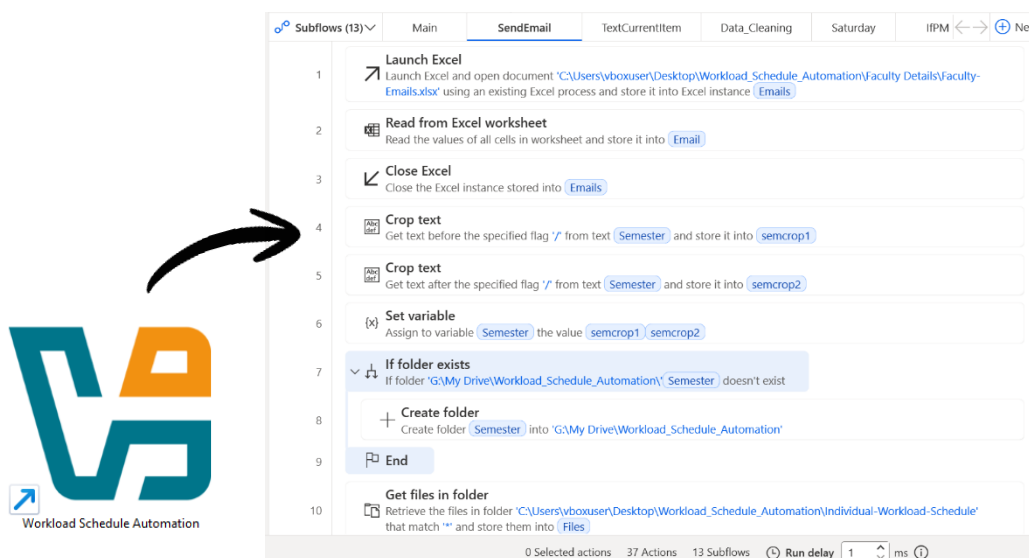
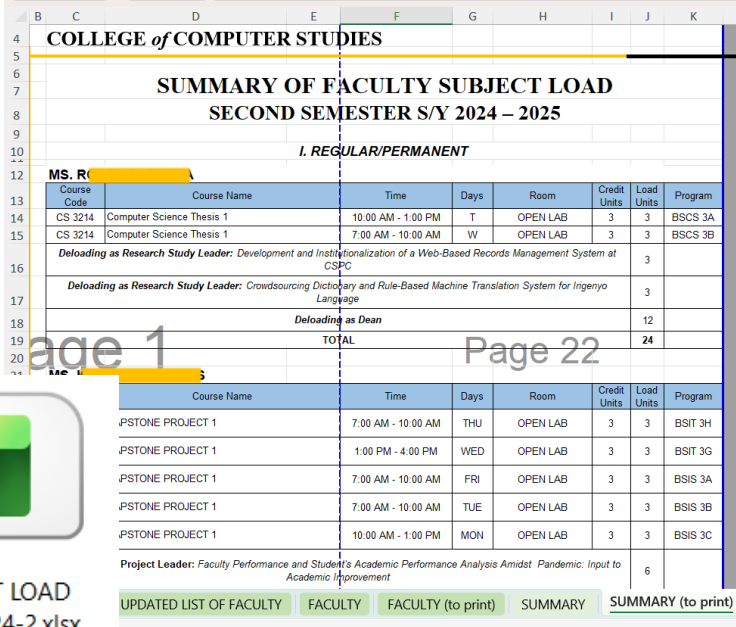


Figure 8. Run Workload Schedule (Desktop) Icon

Figure 8 shows the start of the process named as 'Workload Schedule Automation'. The input of virtual machine were used to help perform the task and to minimize interruption of the workflow. An interaction will begin after clicking the icon set from the desktop, the virtual machine

will launch the .OVA file and will automatically run the automation. This action acts as the trigger of the To-Be Process which will launch the virtual machine and automatically run the automation workflow shortly. This step is done presumably after generating the SLN Excel file as this marks the start of the process where the staff will no longer need to access or open another application inside the virtual machine.



COLLEGE of COMPUTER STUDIES

SUMMARY OF FACULTY SUBJECT LOAD
SECOND SEMESTER S/Y 2024 - 2025

I. REGULAR/PERMANENT

MS. R.

Course Code	Course Name	Time	Days	Room	Credit Units	Load Units	Program
CS 3214	Computer Science Thesis 1	10:00 AM - 1:00 PM	T	OPEN LAB	3	3	BSCS 3A
CS 3214	Computer Science Thesis 1	7:00 AM - 10:00 AM	W	OPEN LAB	3	3	BSCS 3B
	Deloading as Research Study Leader: Development and Institutionalization of a Web-Based Records Management System at CSPC					3	
	Deloading as Research Study Leader: Crowdsourcing Dictionary and Rule-Based Machine Translation System for Iginjo Language					3	
	Deloading as Dean					12	
	TOTAL					24	

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Course Name	Time	Days	Room	Credit Units	Load Units	Program
PSTONE PROJECT 1	7:00 AM - 10:00 AM	THU	OPEN LAB	3	3	BSIT 3H
PSTONE PROJECT 1	1:00 PM - 4:00 PM	WED	OPEN LAB	3	3	BSIT 3G
PSTONE PROJECT 1	7:00 AM - 10:00 AM	FRI	OPEN LAB	3	3	BSIS 3A
PSTONE PROJECT 1	7:00 AM - 10:00 AM	TUE	OPEN LAB	3	3	BSIS 3B
PSTONE PROJECT 1	10:00 AM - 1:00 PM	MON	OPEN LAB	3	3	BSIS 3C
	Project Leader: Faculty Performance and Student's Academic Performance Analysis Amidst Pandemic: Input to Academic Improvement				6	

SUBJECT LOAD NOTICE 24-2.xlsx

UPDATED LIST OF FACULTY | FACULTY | FACULTY (to print) | SUMMARY | SUMMARY (to print)

Figure 9. Verifying File Existence

After generating the Subject Load Notice (SLN) file, the process will require the dean's staff to upload the file named as 'SLN.xlsx' on a Google Drive folder. The folder needs to be addressed in order for the automation to locate the file. Then, the automated robot will check the destination path that points in the google drive containing the SLN Excel file. When the automation detected the file, it interacts with the file and reads the data from the worksheet (shown in Figure 9).

The automation will then extracts data and starts cleaning the document by eliminating all the empty cells and making sure that only the needed information of the faculty is stored. The information is then saved into a variable in PAD named as *ExcelData* and closes the file once all

the data are extracted. The data extraction acts as the basis for the automation to run smoothly as it will no longer need to switch apps from the computer and making sure that the data are already set to fill the data that is prompted as an action for the plotting.

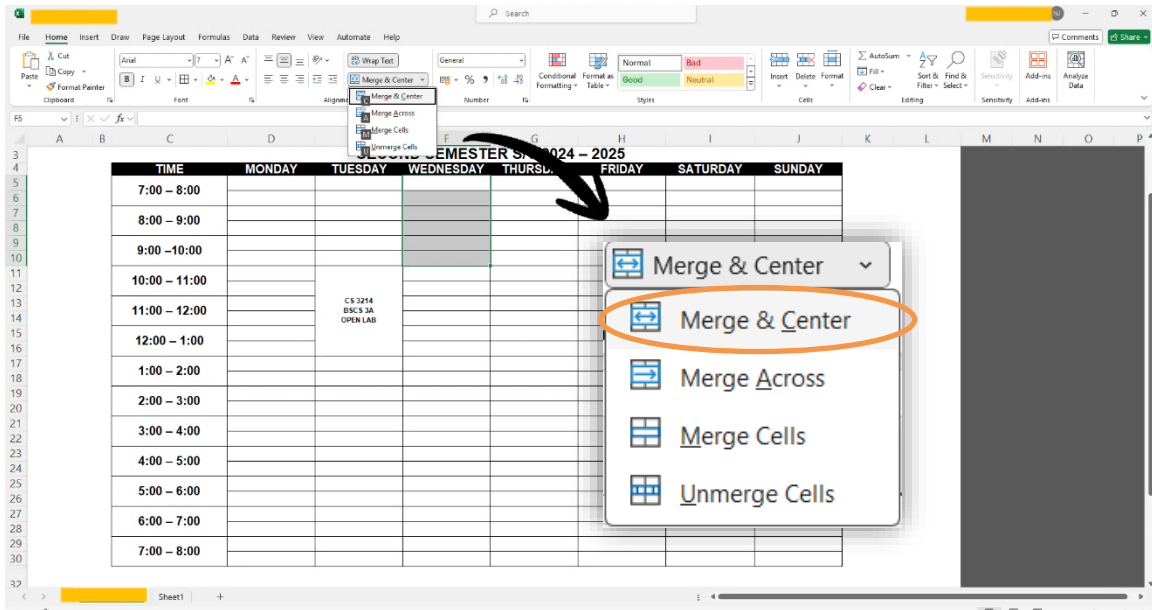


Figure 10. Plotting of Regular and Overload Schedules Using Excel

A workbook template containing the days of the week and other relevant details is launched after storing data from the SLN file. This template is distributed by the dean's staff and were applied in the worksheet for easier plotting of workload schedules. Each information stored by the software robot in PAD's data table will then be plotted at the template as part of the process shown in Figure 10.

The input of plotting the schedule starts with duplicating the *Sheet1* and naming it after the instructor. An interaction is done by selecting a range of cells (that matches the time of each schedule), merge the highlighted cells and then entering details from the *ExcelData* (course code, program and section and the assigned room). Then the automation will input the instructor's name and changing the semester details (e.g *First Semester 2025-2026*). Each time a faculty schedule is generated, it automatically saves a copy in a designated folder for clear-record tracking. After the

conditions of plotting were met, the software robot interacts by deleting the current worksheet and then repeating these actions until all the stored details in the *ExcelData* table are plotted.

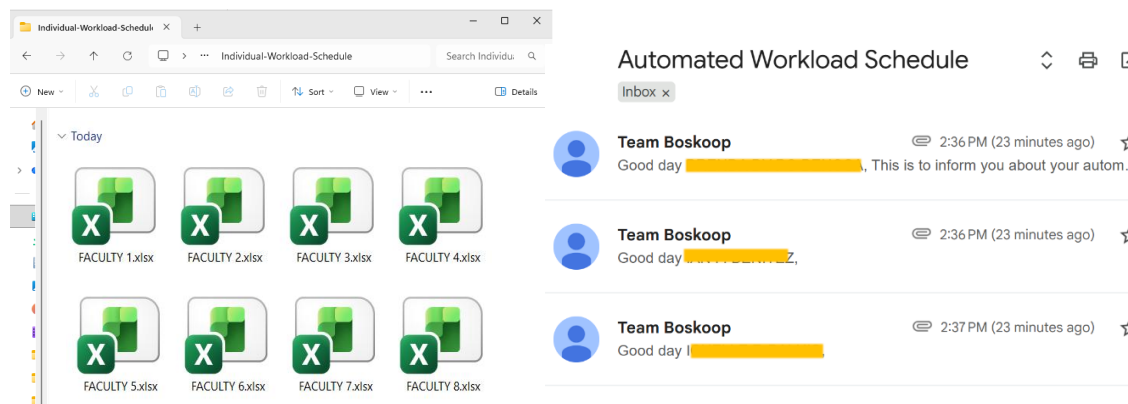


Figure 11. Distribution of File using Email

Figure 11 illustrates the process of saving the complete workload schedule saved in a folder named Individual-Workload Schedule (see *left photo*). This folder contains all the plotted schedule of the faculty which will be sent through each institutional email of the instructor (see *right photo*).

An input using a combination of keyboard-shortcuts and built-in Excel actions from PAD, allowed the procedure to be carried out in keeping the process uniformed and structured. The individual schedules plotted in each worksheet were saved in a folder named Individual-Workload Schedule for distribution of the file to the faculty.

After the complete plotting of the faculty schedules, the automation checks the folder path containing the workbooks named as each instructor. In figure 11, each file is named as ‘Faculty N’ for privacy purposes. These file/worksheets are the plotted schedules generated by the software robot and were saved locally for data storing. The destination path of these workbooks is also stored in a data table in PAD which was set to easily interact with the file without having to interact with other windows from the computer or laptop.

Each file will be sent and is accomplished through the built-in email action feature of Microsoft Power Automate Desktop (PAD). This feature enables the automation to send all the

files without requiring manual intervention from the dean's staff such as accessing or interacting with the email service provider's interface online

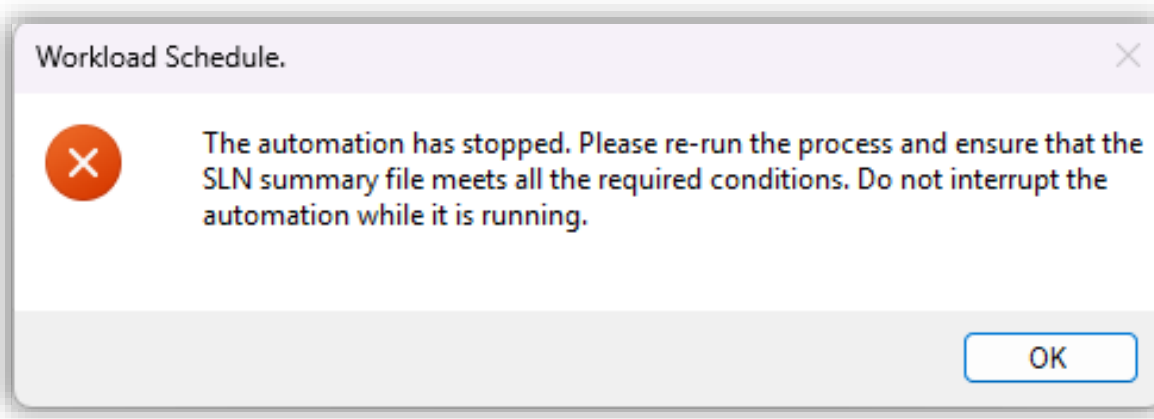


Figure 12. Error Message

Figure 12 shows a dialogue box that appears when the process encounters conditions which do not belong to the configured actions in PAD. An error prompt will appear to inform users that the automation encounters: (1) an unreadable data, (2) interruption to the automation process.

For instance, every time the automation is launched, the user will have to wait for it to run. After the workflow is downloaded and read by the Microsoft PAD, the flow will start by performing the first task set. Often times, during this process, a user apparently navigates between windows which results in an interruption to the automation running. Every time a user interacts while the automation is performing a task, the automation will inform the user of the encounter.

Another reason why the dialogue box appear is when the details that were extracted from the SLN file has date values that were failed to be separated (e.g "HH:MMHH:MM"), this error is often met instead of the correct format "HH:MM–HH:MM". This is why the pop up dialogue box of the automation error appear.

Since the automation is designed to strictly follow each predefined rule, these human actions and errors often lead to failures especially in running case sensitive commands. Minor inconsistencies such as incorrect date formatting can disrupt and triggers an error message.

Good day CCS,

This is to inform you about the Summary Report generated during the run of Workload Schedule Automation.

Best regards,
Team Boskoop

One attachment • Scanned by Gmail ⓘ

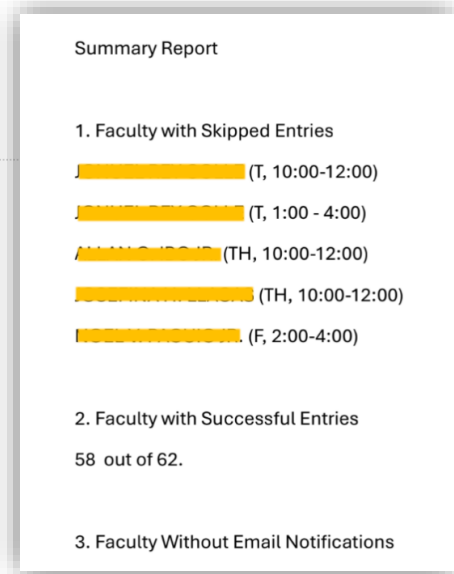
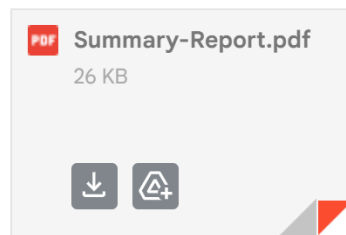


Figure 13. Summary Report

Figure 13 illustrates a summary report that is generated in the process every time the automation encounters overlapping schedules. As seen in the figure, a separate word file is generated and is sent through email with inputs containing the Skipped Entries, Successful Entries, and Faculty without Email Notification, all generated by the software robot.

This report is generated during the launching of PAD where the software robot notes all the conflicts encountered during the interaction of automation flow. Unlike the error message, this summary is generated to guide the dean staff in identifying which schedules from the faculty have existing overlapping issues that made the software robot skip. These conflicts are met based on the version extracted from the SLN file which may or may not consist of human-error-inputs.

Additionally, the total number of successful entries is added in the report to keep track of the number of schedules that were successfully generated. The unsent schedules remain stored in local folder in case the automation has stopped sending mid-action. When the software robot fails to send a copy of schedule to an instructor, the software robot lists the name in the summary report.

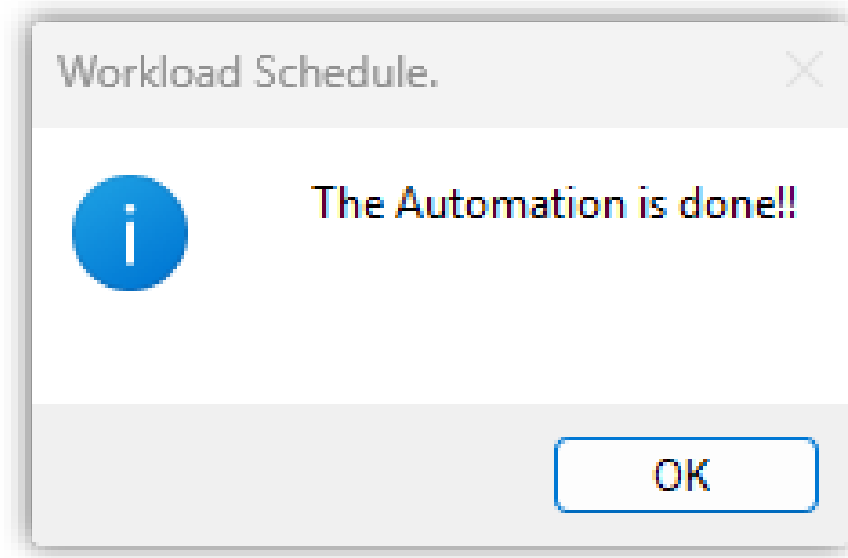


Figure 14. Automation Done

The figure 14 shows a dialogue box that pops out once the automation finishes the process, a message will appear to inform the user that the automation process of inputting each schedule is done. This marks the interaction of automation in successfully executing each action, from plotting the schedule, to sending the worksheets to the designated faculty up until deleting the files (for storage cleaning) marking an end to the automation process. The prompt indicates that all the ruled-based tasks are executed properly and that all logical conditions are met.

The integration of automation using Microsoft PAD were discussed in figures to present a visual flow of how the software robot performs each assigned task. The Hardware Tools were identified to meet the specifications needed of the Software Tools and to accomplish a consistent performance of the automation.

Moreover, the use of virtual machine was added to strengthen the reliability of performing the automation workflow without the need to install or modify the automated-set activities. Overall, the developed automation process shows that the use of Microsoft PAD contributes to a more effective handling of tasks in applying work-related solutions. The use of RPA and virtual machine bridges multiple applications when met with hardware and software capabilities.

4.6 Process Monitoring

During the implementation, the monitoring of the process was conducted to evaluate the performance and effectiveness of automation input to the faculty workload schedules. This phase focused in two key metrics: the accuracy and efficiency, which display the interaction of the automation and the faculty. The logical idea of accuracy refers to the degree to which the automated system produces faculty workload schedules that precisely match the expected and correct outputs in terms of document formatting and day-and-time schedule placement while in conclusion the efficiency revealed comparisons between manual generated tasks and the results on the application of automation.

Moreover, the results were presented using Tables to visually provide the results and compare expected outcomes. While the researchers initially considered achieving 30 target participants using Stratified Sampling, the researchers only achieved ten (10). Four (4) participants from Regular Working Faculty, four (4) from Contract of Service, and two (2) from Part-Timer were sampled. These faculty gave pertinent insights while enabling the practical collecting of data from subjects who were easily accessible [7]. The observation of the results from each participant who participated willingly in the interview were evaluated as monitored and presented in Tables.

4.6.1 Efficiency Results of Automation

A form of table was used to provide a comparison between the duration of manual plotting and the automated version of plotting. The table consists of the recorded time taken to complete each task, as well as the total average time measured in seconds.

The use of automation revealed a positive mark of decreasing the manual total time in plotting schedules to the automated plotting schedule. The measurement revealed the difference between the total time spent in generating faculty workload schedules manually versus the time required to be automated. The table of time efficiency is shown in Table 8.

Table 8: Time Efficiency Evaluation Result

FACULTY	Generated outcome without RPA (seconds)	Generated outcome with RPA (seconds)	Time saved [without-with] (seconds)
Regular Faculty 1	496.8 seconds	43	453.8
Regular Faculty 2	621 seconds	37	584
Regular Faculty 3	249.6 seconds	31	218.6
Regular Faculty 4	435.6 seconds	34	401.6
Contract of Service 1	180 seconds	21	159
Contract of Service 2	143.4 seconds	34	109.4
Contract of Service 3	331.2 seconds	44	287.2
Contract of Service 4	237.5 seconds	34	203.5
Part-Timer 1	88.8 seconds	24	64.8
Part-Timer 2	214.2 seconds	25	203.5
TOTAL	2,998.1 second	327	2,671.1
Average Time	299.8 second	32.7	267.1

The table above shows the results of each manual and automated process of the workload plotting process. The average time were measured by dividing each generated outcomes and time-saved to the ten faculty members.

To measure the desired outcomes, each faculty were assigned to plot their assigned schedules provided in the previous semester. Roughly around 2998.1 seconds (49.97 in minutes) were achieved in plotting the schedules manually, with an average of 299.8 seconds (4.9 minutes) showing the usual time it takes for ten faculty members to finish the plotting process. In the integration of RPA, the software robot achieved a total of 327 seconds (5.45 minutes) with an average of 32.7 seconds to generate schedules of ten faculty members. A total of 2671.1 seconds (45 minutes) were saved with an average time of 267.1 seconds (4 minutes). This revealed that the use of RPA 1.0, provided 89% faster approach of using software robot in the process.

The time-efficiency table presented in Table 8 recognizes the results of each task in both the faculty and the performance of the software robot. The results revealed a significant improvement in processing the time of generating the faculty workloads. Additionally, the software

robot has proven that automation greatly enhances the workflow especially the tasks that are done repetitively by the faculty.

4.6.2 Accuracy Results of Automation

To assess the accuracy of the implemented solutions and establish clear connection to the efficiency results the automation were evaluated by the dean's staff. In this study the accuracy were measured by inputting '1' as accurate and '0' as inaccurate in each faculty designation. The scoring will be based on Document Format and Schedule (Day and Time) to provide basis and evaluation on how the software robot performs. Additionally, the accuracy results help validate the efficiency on outcomes generated by RPA. This achieves the purpose of the study and the objectives of performing the desired results. Thus, the Accuracy Evaluation Result is presented to prove how the system performed the work intended and is presented in Table 9.

Table 9: Accuracy Evaluation Result

FACULTY	Document Format	Schedule (Day and Time)	Accuracy
Regular Faculty 1	1	1	100%
Regular Faculty 2	1	1	100%
Regular Faculty 3	1	1	100%
Regular Faculty 4	1	1	100%
COS 1	1	1	100%
COS 2	1	1	100%
COS 3	1	1	100%
COS 4	1	1	100%
Part-Timer 1	1	1	100%
Part-Timer 2	1	1	100%
Total:			100%

$$\text{Accuracy} = [1/10(1+1+1+1+1+1+1+1+1+1)] * 100 = 100\%$$

Table 9 presented above assesses the ability of the software robot to accurately interact in the process by producing faculty workload outputs based on the three columns: Document Format, Time, and Schedule. These columns were used to evaluate whether the automated faculty workload

is correctly interpreted, processed and plotted based from the provided Subject Load Notice (SLN) file.

The result reveals that the automated faculty workload achieved a 100% accuracy rate across all ten-faculty member. The generated workload schedule of each faculty member demonstrates that the use of RPA-PAD were fully functional and shows an accurate formatting, correctness to time allotment and schedule. The faculty's feedback on the accuracy of the automated response was discussed as follows.

Regular Faculty 1 and 2 responded with satisfaction that the automation achieved a 100% accuracy in terms of document format, time allotment and schedule plotting. Regular Faculty 2 said that the RPA successfully interpreted the SLN which took almost 621 seconds to plot. Additionally, Instructor 1 confirms that RPA's ability to correctly assign teaching hours and subject details are accurately faster and efficient.

Part-Timer 1 and 2, COS Instructor 1 and 3, responded positive feedback as the results generated using automation showed a 100% accuracy. The performance of automation ensured that the details provided in the SLN matched the plotted data using automation. The accuracy were also believed to be beneficial to the instructors as they will no longer need to plot schedules manually in terms of their teaching hours.

The outcome generated proved the effectiveness of RPA to Regular 3, and COS Instructor 4, leading to a satisfied performance of the automation. The instructors commended the automation's ability to time plotting and data accuracy as it accurately plotted the details, including subject codes, class times and assigned rooms.

Moreover, the result were criticized by Regular Faculty 3 and 4 pertaining that the additional schedules should be added. However, the integration of automation were limited to the source file (SLN), on which additional schedules aren't indicated. Having criticized, the process achieved a 100% accuracy to the result of automation.

COS Instructor 2 also generated 100% accuracy in terms of plotting detailed schedule integrated with automation. The instructor suggested a minimal fix in the workload template, but still leaving a satisfied and positive feedback to the result.

In Summary, the accuracy helped reveal precisions covering the document format and the schedule that were scored across all categories. The software robot achieved a 100% result and consistency across regular faculty, contract of service personnel, and part-time instructors indicating that the automated system can produce the schedule formats without errors.

To further reinforce and support the findings, an inspection of the data were done to validate the researchers' hypotheses [8]. The values determined were based on the outcomes performed by the software robot. These values were presented in Table 10.

Table 10: Accuracy Inspection Data

METRIC	Value
Total Samples	10
Correct Outcome	10
Incorrect Outcome	0
Accuracy	100

Using the formula introduced in Chapter 3, all ten (10) schedules in the samples have been verified in the process to display the required correct outcomes. Since no errors were detected in any of the inspected schedules, the incorrect outcome was marked and inputted as zero (0). These results proved that by implying correct actions in the PAD, and by identifying correct interaction of workflow in the process, the use of RPA 1.0 using PAD, can achieve the best accuracy of results.

In conclusion, the evaluation of the efficiency and accuracy as presented in Tables, revealed a positive impact from the faculty as the performance of software robots were measured to effectively minimize the time while maximizing accurate results. The accuracy evaluation measures revealed that the generated faculty workload schedules by the software robot are correct and reliable, backed by accuracy inspection of the data, the results supported to provide an correct

outcomes when using RPA. Meanwhile, the effectiveness of automation using time-efficiency evaluation result showed a significant amount of time reduced in completing tasks and in identifying the importance of each average time that were saved

Notes

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CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the results and key findings of the study on Faculty Workload Schedule Generation in the College of Computer Studies. It includes the summary, findings, conclusions and recommendations.

5.1 Summary

Employees, as seen in the modern world, are exposed to doing office operations thanks to various technological advancements that are made to enhance the ability to handle operational processes. While these may seem to be a perfect example of innovation, studies show that managing such operations in an office environment requires efficient planning and implementation of techs to meet sleek office operations and better work outcomes.

The College of Computer Studies in the Camarines Sur Polytechnic Colleges is an institution that execute tech-driven approaches in its operations. But since the world is an ever-changing cycle of progress and developments, current procedures handled in the college that were once efficient, have impacted a reasonable dissatisfaction to complying college requirements. Outdated management to these office practices has stacked in an endless loop of monotonous compliance that are often complained by the faculty to be time-consuming especially when mishandled.

The Integration of Robotic Process Automation (RPA) to the Academic Operational Tasks of the College of Computer Studies: A Module Focusing on the Generation of Faculty Workload Schedule is an undergraduate study aiming to provide operational enhancements to the college. This study seeks to enhance faculty workload generation process by recognizing the current procedures and determining its automation potential while measuring its precision and effectiveness of putting RPA 1.0 into practice.

The use of multimethod approach allowed to achieve the three (3) identified objectives of this study. The qualitative method used by the researchers consists of semi-structured interviews and observations to collect several possible patterns related to plotting the faculty schedule, while the quantitative approaches allowed for numerical interpretation using formulas to scale, simulation of RPA and inspection of data, which also help identify the degree of the tasks. The multimethod approach allowed the researchers to use both approaches interdependently to interpret each collected data in the Business Process Management (BPM) Framework.

To perform comprehensive analysis of the process of plotting the schedule of faculty, this study adopts the BPM Framework as its guiding step. A graphical language called Business Process Model Notation (BPMN) helps to discuss guided-blueprint in identifying, re-designing, and produce automated-solutions before implementing the automated system. In the first phase of the BPMN framework: Discovery, the researchers identified the current set up that are executed in the college. The As-Is Process Model helps to visually graph and provide an overview of the entities involved in the practices related to the plotting of each workload schedule. These tasks were assessed to follow the second Phase of the Framework: Analysis. Using Multi-Criteria Scoring Model and Value-Complexity Matrix, the researchers were able to evaluate the automatability of the tasks recognized, and its complexity. The third Phase of the Framework: Redesign, were mapped out as To-Be Process Model, displaying an overview of the proposed automated solution and the technicality of how the software robot performs. The fourth Phase: Implementation, were carried out to test the software robot in actual operations in the process of generating the schedules. After achieving the desired results, the final Phase of the Framework: Monitoring, were achieved by showing a huge amount of time (a total of 2671.1 seconds (45 minutes)) saved. These Evaluation Result allowed the researchers to measure the performance of the applying RPA and provide an evidence-based result in supporting the findings.

5.2 Findings

Following the analysis of the data collected, the researchers led to the following findings:

1. The existing process in the College of Computer Studies reveal tasks involved in creating the workload schedule of the faculty. These tasks were assessed by the researchers using the Multi-Criteria Scoring Model and Value-Complexity Matrix to determine which task should be considered as candidate for automation. On Multi-Criteria Scoring Model, the tasks were categorized into levels of their automatability based on the criteria including the process structure, level of standardization, repetitiveness, degree of user involvement, and task volume. In a total nineteen (19) tasks identified, seventeen (17) of these tasks scored a 'High' degree while two (2) tasks were identified to be 'Medium'. Based on the results, all tasks can be automated, garnering a percentage of 89.47% being 'High' while 10.53% being 'Medium'. The Value-Complexity Matrix also supported the result of tasks that were too complicated for the automation to perform, these four (4) tasks show a 'High' Value while also indicating 'High' Complexity in terms of processing. The four (4) tasks were identified to be the task related to inputting additional schedules (Research, Quasi-teaching, Consultation and the Highlighting of Overload Schedule) that weren't indicated in the SLN file.
2. The workload schedule generation process was redesigned using Business Process Management Framework (BPM) to provide a To-Be overview and technical graph on how the automation will perform in the redesigned process model. The automated workflow includes the clicking of a desktop icon to run the automation, extracting and cleaning the data from the SLN Excel file, plotting of schedules, and sending the generated workload schedule file to each faculty institutional emails. The use virtual machine were enforced to easily perform the automation using Power Automate Desktop that was used to integrate Robotic Process Automation (RPA) 1.0.
3. The Integration of RPA 1.0 into the workload schedule generation process were monitored using a set of formula in assessing the accuracy and calculate the time-saved in applying RPA. The Time-

Efficiency Evaluation Result were used to display each outcome both manually and using the software robot, revealing an average of 300 seconds were recorded while with RPA integration, it was reduced to only 37 seconds. Moreover, the Accuracy Evaluation Result were used to assess the format correctness, and the schedules plotted by the software robot. With ten (10) documents generated, the results revealed an accuracy of 100% with 10 correct outcomes generated.

5.3 Conclusions

The following conclusions were drawn based on the findings of the study.

1. The manual process of generating the faculty workload schedule was examined with the help of Automatability Scoresheet that revealed the automatability of each task. The results display a high remark in the criteria which indicates to be automatable. Meanwhile the use of Value-Complexity Matrix helps prioritize tasks that display a high value in the process and reveal tasks that are highly complex for a software robot to perform.
2. The integration of RPA were graphed first using the BPMN, which helped the researchers to create the To-Be Process Model that graphically map visuals on how the redesigned process works. The integration of RPA to the identified tasks were deployed using virtual machine tool (VirtualBox) which helps to easily deploy automation on the recipient of this study.
3. The automation made a measurable performance, which consistently generated a workload schedule with 100% accuracy. The evaluation measures revealed that by applying this automated solution, a reduce in the overall processing time by more than 89% is achieved. These results highlight the potential of RPA 1.0 as a reliable solution in generating the faculty workload schedule and in improving the operational efficiency within the College of Computer Studies (CCS).

5.4 Recommendations

In light of the conclusions, the following recommendations were identified:

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1. Future researchers may improve this study by integrating more advanced version of RPA such as RPA 2.0 or RPA 3.0 or by using other automation tool aside from Microsoft Power Automate Desktop (PAD).
 2. Future researchers can extend the integration of automation to other college departments of the Camarines Sur Polytechnic Colleges (CSPC).
 3. The flow control logic of the automation process may be improved by future researchers by adding the schedules not indicated in the SLN file.
 4. Future researchers may explore other methodology other than Business Process Management (BPM) Methodology to further enhance scalability and efficiency while aligning the objectives and scope to their study.

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APPENDICES

APPENDIX A

EVALUATION TOOL

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED

COLLEGE of
COMPUTER STUDIES

Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score


Ms. Maria Daisy Belardo
Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
	2:39	34 secs.

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
2:39	34 secs.	1:82		

Observations:

workload for CDS & Admin Part-time should be in PDF format.


Ms. Maria Daisy Belardo

Faculty



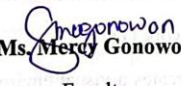
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	


Ms. Mercy Gonowon

Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
	3.15	34 secs.

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
3.15	34 secs.	3.61		

Observations:


Ms. Mercy Gonowon
Faculty



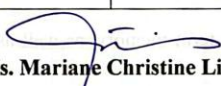
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score


Ms. Mariane Christine Lico

Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
Cleaning of Template Plifting	1:48	24 sec.

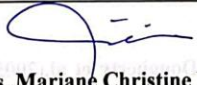
Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
1:48	24	1:24		

Observations:


Ms. Mariane Christine Lico

Faculty



Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	1


Ms. Maivie Ramizares

Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
Plotting	3:52	44 secs.

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved (%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
3:52	44 secs.	4.78		

Observations:


Ms. Marivic Ramizares

Faculty



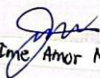
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score


Ms. Ine Amor Montel

Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
	4:16	31 secs.

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
4:16	31 secs.	3:64		

Observations:

Can include info on overload, lecture / lab , etc.

Ms. Ime Amor Monte

Faculty



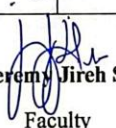
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	


Mr. Jeremy Jireh S. Neo
Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
plotting	10:35	37 secs

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
10:35	37 secs.	9:73		

Observations:


Mr. Jeremy Jireh S. Neo

Faculty



Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	1

Ms. Jillian Lourdes Orante

Faculty

Republic of the Philippines
CAMARINES SUR
POLYTECHNIC COLLEGES
ISO 9001:2015 CERTIFIED



COLLEGE of
COMPUTER STUDIES

Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
Cleaning of Document Plotting	3:00	21 sec.

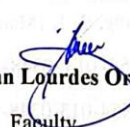
Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
3:00	21 sec.	2:65		

Observations:


Ms. Jillian Lourdes Orante
Faculty



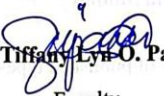
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	


Ms. Tiffany Lynn O. Pandes
Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
	7:26	34 secs.

Simulation Run Log

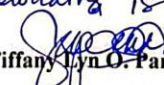
Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
7:26	34 secs.	6.69		

Observations:

No overload identification,
Other activities such as extension, research,
quasi-teaching, consultation hours to students, debloating
at PC or other debloating is not reflected.


Ms. Tiffany Lynn O. Pandes
Faculty



Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	


Mr. James Nicolo Sias
Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA : Time (mins)
	8 : 28	43 secs

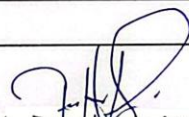
Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

Average Manual Time (mins)	Average RPA Time (mins)	Time Saved (%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
8 : 28	43 secs.	7 : 56		

Observations:


Mr. James Nicolo Sias
Faculty



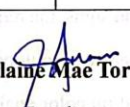
Inspection Journal

Table Format for Recording Observations

The goal of this study is to improve data accuracy in academic processes by integrating robotic process automation (RPA) into the College of Computer Studies (CCS). To identify the accuracy of the automation, the data were extracted from the Subject Load Notice (SLN) to ensure correctness and reliability. Any data extraction error or automation result delay shall receive a score of zero (0), whereas an RPA procedure that accomplishes optimal efficiency and precise data extraction is given the highest possible score of one (1).

Table Format for Accuracy Checking

Document Format	Time	Schedule	Score
1	1	1	1


Ms. Elaine Mae Tormes
Faculty



Simulation Journal with Test Cases

Task Name	Manual Time (mins)	RPA Time (mins)
	3:57	25 secs

Simulation Run Log

Task Name	Run	Start Time	End Time	Total Time (mins)	Time Saved	CPU Usage (%)	Memory Used (MB)	Notes

Summary

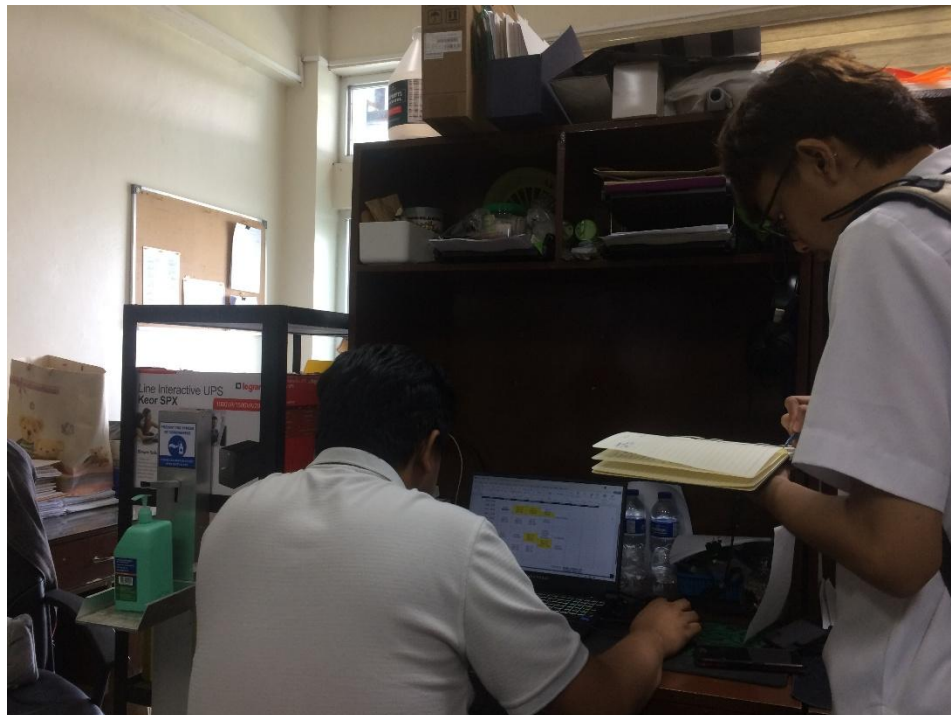
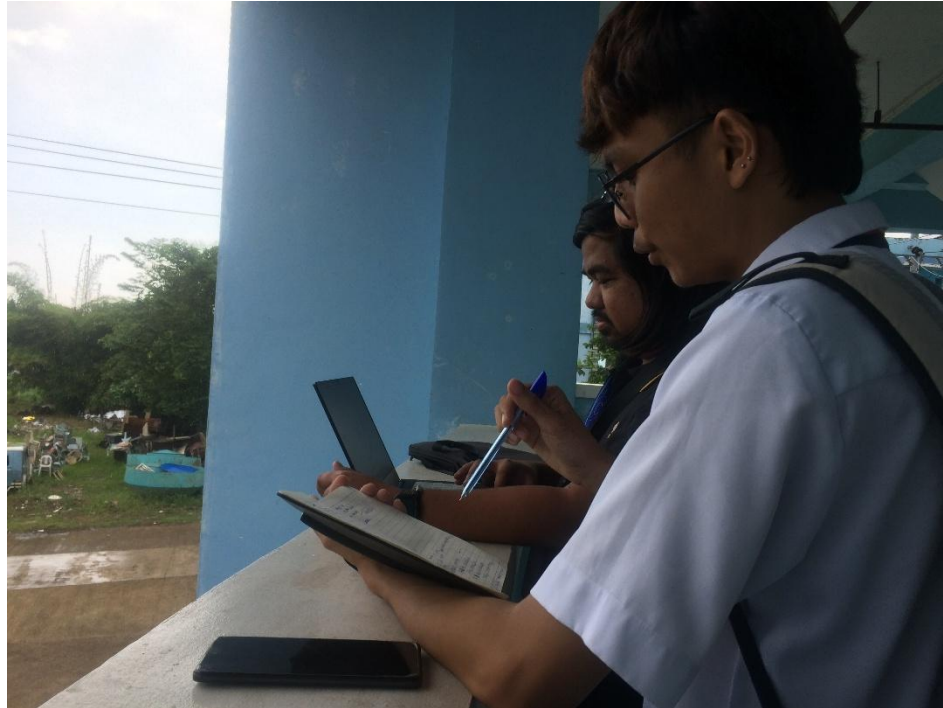
Average Manual Time (mins)	Average RPA Time (mins)	Time Saved(%)	Avg CPU Usage (%)	Avg Memory Usage (MB)
3:57	25secs.	3:35		

Observations:


Ms. Elaine Mae Tormes
Faculty

APPENDIX B

DOCUMENTATION







APPENDIX C

SAMPLE INPUT/OUTPUT/REPORTS

WORKLOAD SCHEDULE OF FACULTY
FIRST SEMESTER S/Y 2025 – 2026

File Code: 1.7.1

TIME	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
7:00 – 8:00		ITA 213 BSIT 2D ACAD1-103		CGIT 106 BSIT 3H MAC Laboratory	CGIT 106 BSIT 3H ACAD1-103		
8:00 – 9:00							
9:00 – 10:00							
10:00 – 11:00							
11:00 – 12:00		ITA 213 BSIT 2D MAC Laboratory	ITA 213 BSIT 2C MAC Laboratory		ITA 213 BSIT 2A Room 2		
12:00 – 1:00							
1:00 – 2:00	ITA 213 BSIT 2C Room 6						
2:00 – 3:00			CGIT 106 BSIT 3E ACAD1-103	ITA 213 BSIT 2A MAC Laboratory	CGIT 106 BSIT 3F MAC Laboratory		
3:00 – 4:00							
4:00 – 5:00		CGIT 106 BSIT 3F Room 3					
5:00 – 6:00	CGIT 106 BSIT 3E MAC Laboratory						



Automated Workload Schedule Inbox x



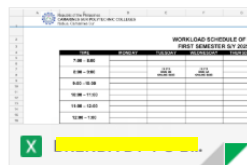
Team Boskoop [redacted] via cloudflare-smtp.org
to dummy77-639214 ▾

Good day [redacted]

This is to inform you about your automated workload schedule for the FIRST SEMESTER S/Y 2025 – 2026.
Please review the attached file for details.

Best regards,
Team Boskoop

One attachment • Scanned by Gmail Add to Drive



APPENDIX D

USER'S GUIDE

Standard Operating Procedures (SOP) in Utilizing Robotic Process Automation (RPA) 1.0 in Generation of Workload Schedule of Faculty

A Capstone Project Presented
In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Systems

Prepared by:

Adrian G. Saballero
Project Head

Niño Jaybee R. Bermas
Information Manager

Ruby Mae O. Penolio
Manuscript

Reviewed by:


Shiela Dona S. Manlapaz
Adviser


Jayvee Niel S. Sias
Consultant

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GUIDELINES AND PROCEDURES	5
1. Preparation Phase	5
2. Executing RPA 1.0 Automation	6
ERROR HANDLING AND TROUBLESHOOTING	7
REVISION HYSTORY	8

I. PURPOSE

Standard Operating Procedures' purpose is to give a clear guide on how to use the Robotic Process Automation (RPA) 1.0 in Generating the Workload Schedule of Faculty. This is made to improve the current manual process by improving accuracy and efficiency while reducing error and saving time for the faculty members.

II. SCOPE

This Standard Operating Procedure is intended for the College of Computer Studies and covers all the steps needed in operating the RPA tool properly. It includes file preparation and running the automation. However, for the Regular Faculty members, this SOP does not include additional schedule such as overload, quasi-teaching, research, extension and de-loading.

III. DEFINITION OF TERMS

To provide better understanding and to avoid problems in understanding the terms used throughout the SOP, the following technical terms are conceptually and operationally defined.

- a. RPA 1.0** - A technology that uses software robots to automate repetitive task by copying human interactions with digital systems. This is a system used to automate the process of generating the workload schedule of the faculty in the College of Computer Studies.
- b. Subject Load Notice (SLN)** – This is a document created by the College of Computer studies that contains the faculty teaching load for a specific semester. It includes essential details such as faculty name, school year and semester, subject code,

schedules and room. This serves as primary data source for generating each faculty workload schedule.

- c. **Workload Schedule Template** – This is an ISO registered document used by the college to formally present the teaching schedule of faculty. The Software Robot automatically populate this template to produce accurate and consistent workload schedule.
- d. **Software Robot** – A programmed virtual worker that perform digital task by following predefined rules and instructions.
- e. **Power Automate Desktop (PAD)** – This is a low code automation tool developed by Microsoft that enables user to create a software robot capable of automating repetitive and rule-based task. This serves as primary platform for developing and executing the automated workflow for generating the workload schedule of faculty.
- f. **Virtual Machine** – A software computing environment that act as separate computer system within a physical device. This allows the automation to run independently in background allowing user to continue using the host computer without interrupting the RPA operation.

IV. ROLE AND RESPONSIBILITIES

- a. **Dean's Staff** – The Dean's Staff serves as primary operator, responsible for preparing and validating data in SLN before it is used in automation. They are the one that will upload the SLN file to the designated GDrive folder and run the automation in the Virtual Machine.
- b. **Faculty** – The Faculty member act as secondary validator who review their automated workload schedule. They are responsible for confirming the details

such as time, course code, room. If there is concern about the schedule, it must be reported to the Dean's Office for correction of their schedule in SLN.

V. GUIDELINES AND PROCEDURES

1. Preparation Phase

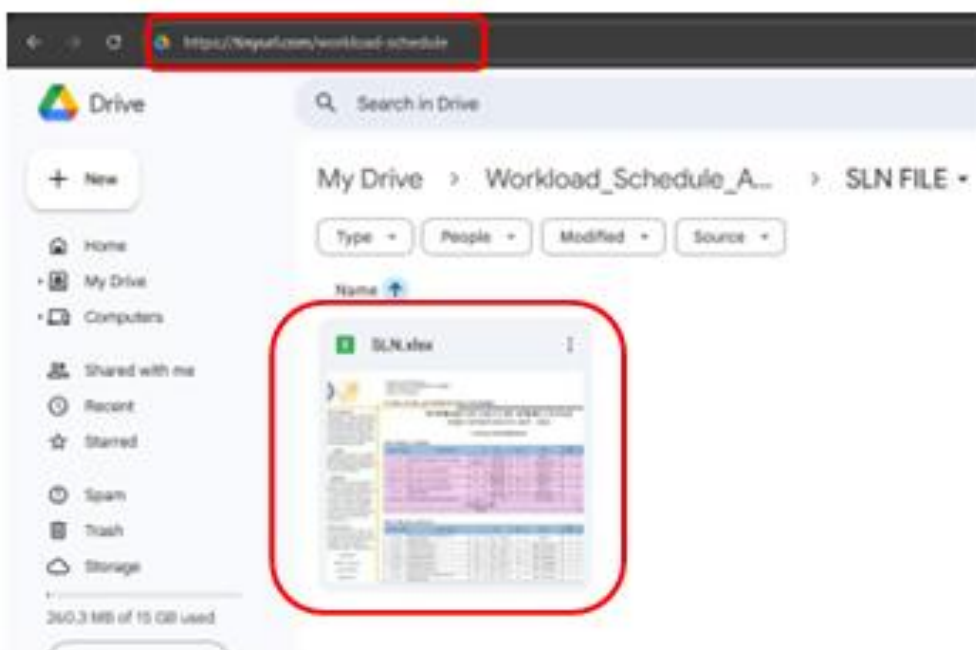
Step 1. Get the latest Subject Load Notice document from Dean's Office.

Step 2. Perform Validation Check for the following:

- Correct Semester and Academic Year
- Accurate Faculty Names
- Accurate day and room assignment
- Correct Time Format HH:MM – HH:MM (e.g. 10:00AM – 1:00PM)

Step 3. Save the Subject Load Notice document with the filename "SLN.xlsx".

Step 4. Upload the "SLN.xlsx" file to GDrive Folder Link "<https://tinyurl.com/workload-schedule>".



2. Executing RPA 1.0 Automation

Step 5. Launch the Virtual Machine icon Named “Workload Schedule Automation”.



Step 6. Run the Automation through Python Script or Power Automate Console.

a. Run Automation through Python Script

Step 6. Click the icon in the Desktop named “PythonScript”.



This will run the automation automatically.

b. Run the Automation through Power Automate Console

Step 6. Open Power Automate Desktop.

Step 7. Select Flows.

Step 8. Select the Flow named “Workload_Schedule_Automation”.

Step 9. Click Run.



VI. ERROR HANDLING AND TROUBLESHOOTING

- a. Incorrect Time Format** – The SLN File may contain inconsistencies such as overlapping time, incorrect formatting of time (e.g., “10:00AM1:00PM” instead of “10:00AM - 1:00PM”). The software bot relies heavily on standardized format. If this issue persists, recheck the SLN file and ensure all the time in schedule follow the time format “HH:MM-HH:MM”, and re-run the automation.
- b. Virtual Machine Authentication** – The Virtual Machine (VM) may require manual authentication due to session timeout, system updates or security policy that automatically log-out inactive users. If this issue persists, enter the assigned system password “qwerty1234” to unlock the VM and restore access.
- c. Slow Internet/No Internet Connection** – During automation, if the internet connection suddenly slow down or disconnected completely, the automation can continue to plot the workload schedule of faculty offline. Any steps that

require internet connection such as sending to email will fail and trigger an error. If this issue persists, check the stability of the internet connection. Once the connection is restored, re-run the automation to allow the software bot to send the faculty workload schedule to their email and complete the process.

VII. REVISION HYSTORY

Version	Date	Description of Changes	Prepared By	Approves By
1.0	11/12/2025	Initial Release of SOP	Team Boskoop	Adviser

APPENDIX E

COMMUNICATION LETTERS



ISO 9001:2015

Republic of the Philippines
Camarines Sur Polytechnic
Colleges



COLLEGE of COMPUTER STUDIES

August 12, 2025

MS. ROSEL O. ONESA

OIC Dean, College of Computer Studies
This College

Dear Ma'am,

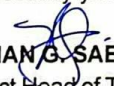
We are Team Boskoop, from BSIS 4B, currently working on our capstone project entitled: **"Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty"**. The purpose of this capstone project is to incorporate RPA in the process of creating a *faculty workload planner*.

In light with this, we formally request permission to allow us to conduct an interview and observation in the Faculty and Deans Office. This will help us gain enough insights to identify how the current way (of creating workload planner) is handled individually. With appropriate permission to facilitate the interview, we take that this will be beneficial for the purpose of our project in the college.

We also assure you that the interview will be treated with confidentiality and will be used solely for academic approach in this college.

Thank you for your time and consideration. We look forward to your positive response.

Respectfully yours,


ADRIAN G. SABALLERO
Project Head of Team Boskoop

Noted By:


ICHELLE F. BALUIS, MSIT
Capstone Project Subject Adviser

Approved by:


ROSEL O. ONESA, MIT
OIC Dean, College of Computer Studies



Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified

COLLEGE of COMPUTER STUDIES

QUALITY POLICY

CSPC commits to provide the highest standards in total customer satisfaction in accord with legal and other requirements to satisfy current and future needs of clients and stakeholders; uphold good governance conditions; and sustain the integrity of the environment and its resources through continual improvement of its quality and environmental management system.

VISION

CSPC envisions a dynamic, inclusive, resilient and globally competitive polytechnic education institution committed to achieving excellence and advancing innovation that transcends societies in the Bicol region and beyond.

MISSION

CSPC is highly committed to delivering quality polytechnic education, providing excellent services, and promoting sustainable development and resilience in alignment with legal and statutory requirements, through cutting-edge technological innovation, impactful extension services, robust research culture, efficient resource management, and mutually beneficial partnerships with local and international stakeholders.

CORE VALUES

CSPC shall pursue its mission and vision with fervor steered by the values that have driven this institution over the years. Our services and our acts shall constantly reflect these ten organizational values that CSPC TRASCENDS:

Teamwork
Responsiveness
Accountability
Nurturance
Commitment
Excellence
Networking Synergy
Determination
Social responsibility

GOALS AND OBJECTIVES

- **Continued Academic Excellence.** To produce future-ready graduates with professional and social competencies.
- **Strengthened Research and Innovation.** To prioritize the creation of new knowledge and technology through globally responsive and socially relevant research practices towards measurable social progress.
- **Engaged Sustainable Communities.** To engage with local communities in enhancing standards of living as anchored to the Ambisyon Natin 2040 and the UN Sustainable Development Goals.
- **Balanced Finances through Production and Resource Generation.** To capitalize on innovative and self-sustaining business ventures.
- **Optimized Digitalization for Good Governance and Operational**

August 27, 2025

MS. ROSEL O. ONESA
OIC Dean, College of Computer Studies
This College

Dear Ma'am,

Greetings!

We are Team Boskoop, from BSIS 4B, currently working on our capstone project entitled: **"Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty"**. The capstone project will focus on automating the process of creating a faculty workload by integrating RPA into the college.

In light with this, we send this letter to issue *a formal request to ask for an available time from your Admin staff* to explore the purpose of integrating RPA solutions intended for automating the workload scheduling plan of the faculty. With appropriate permission to facilitate the development and testing of the said inquiry, this will shed light on the purpose of our study.

We assure you that any data will be treated with confidentiality and used solely for academic purposes in this college

Thank you for your time and consideration. We look forward to your positive response.

Respectfully yours,


ADRIAN C. SABALLERO
Project Head of Team Boskoop

Noted By:


ICHELLE F. BALUIS, MIT
Capstone Subject Adviser

Approved by:


ROSEL O. ONESA, MIT
OIC Dean, College of Computer Studies



Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified

COLLEGE of COMPUTER STUDIES

QUALITY POLICY

CSPC commits to provide the highest standards in total customer satisfaction in accord with legal and other requirements to satisfy current and future needs of clients and stakeholders; uphold good governance conditions; and sustain the integrity of the environment and its resources through continual improvement of its quality and environmental management system.

VISION

CSPC envisions a dynamic, inclusive, resilient and globally competitive polytechnic education institution committed to achieving excellence and advancing innovation that transcends societies in the Bicol region and beyond.

MISSION

CSPC is highly committed to delivering quality polytechnic education, providing excellent services, and promoting sustainable development and resilience in alignment with legal and statutory requirements, through cutting-edge technological innovation. Impactful extension services, robust research culture, efficient resource management, and mutually beneficial partnerships with local and international stakeholders.

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CSPC shall pursue its mission and vision with fervor steered by the values that have driven this institution over the years. Our services and our acts shall constantly reflect these ten organizational values that CSPC TRANSCENDS:

Teamwork
Responsiveness
Accountability
Nurturance
Commitment
Excellence
Networking Synergy
Determination
Social responsibility

GOALS AND OBJECTIVES

- Continued Academic Excellence. To produce future-ready graduates with professional and social competencies.
- Strengthened Research and Innovation. To prioritize the creation of new knowledge and technology through globally responsive and socially relevant research practices towards measurable social progress.
- Engaged Sustainable Communities. To engage with local communities in enhancing standards of living as anchored to the Ambisyon Natin 2040 and the UN Sustainable Development Goals.
- Balanced Finances through Production and Resource Generation. To capitalize on innovative and self-sustaining business ventures.
- Optimized Digitalization for Good Governance and Operational Excellence. To increase organizational productivity and performance through digital migration, good governance principles and sound management techniques towards enduring economic prosperity.

October 23, 2025

MS. ROSEL O. ONESA

OIC Dean, College of Computer Studies
This College

Dear Ma'am,

Greetings!

We are Team Boskoop, from BSIS 4B, currently working on our capstone project entitled: **"Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty"**. A capstone project intended to focus on applying automation to the process of creating a *faculty workload schedule*.

In light with this, we send this letter as a *formal request to ask for an available time from your Admin staff* to explore the purpose of integrating the RPA solution. Moreover, this letter seeks to request permission to use the college email *guided by your staff* to simulate and implement our capstone output. With appropriate permission to facilitate the development and testing of the said inquiry, this will shed light on the purpose of our study.

We assure you that any data will be treated with confidentiality and used solely for academic purposes in this college

Thank you for your time and consideration. We look forward to your positive response.

Respectfully yours,


ADRIAN G. SABALLERO
Project Head of Team Boskoop

Noted By:


ICHELLE F. BALUIS, DIT
Capstone Subject Adviser

Approved by:


ROSEL O. ONESA, MIT
OIC Dean, College of Computer Studies

APPENDIX F

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: February 12, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Shiela Dona S. Manlapaz, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

SHIELA DONA S. MANLAPAZ, MIT

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) in streamlining an Instructor's Workload on the College of Computer Studies

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

- a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


SHEILA DIGNA S. MANLAPAZ, MIT
Expert

DATE: FEBRUARY 12, 2025


ROSEL O. ONESA, MIT
CSPC Representative

DATE: FEBRUARY 12, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: February 12, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Jayvee Niel S. Sias hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

JAYVEE NIEL S. SIAS

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) in streamlining an Instructor's Workload on the College of Computer Studies.

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

- a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


JAYVEE NIEL S. SIAS
Expert

DATE: FEBRUARY 12, 2025


ROSEL G. ONESA, MIT
CSPC Representative

DATE: FEBRUARY 12, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: February 12, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Khari Vincent Requinala hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

KHARL VINCENT REQUINALA

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) in streamlining an Instructor's Workload on the College of Computer Studies.

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
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 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

- a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


KHARL VINCENT REQUINALA
Expert

DATE: FEBRUARY 12, 2025


ROSEL O. ONESA, MIT
CSPC Representative

DATE: FEBRUARY 12, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 21, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Brenda D. Benosa, MIS hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

BRENDA D. BENOSA, MIS

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:

- a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


BRENDA D. BENOSA, MIS
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: May 21, 2025

DATE: May 21, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 21, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Jonuel Rey Colle hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

JONUEL REY COLLE

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
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 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
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- a. which is or becomes publicly available without breach of this Agreement;
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8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
- a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
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11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


JONUEL REY COLLE
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: May 21, 2025

DATE: May 21, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: May 21, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Noel V. Paguio Jr. hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:

NOEL V. PAGUIO JR.

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:

ROSEL O. ONESA, MIT

2. The confidential information disclosed under this Agreement is described as:

Contents of the TCP by:

Team Boskoop: Niño Jaybee R. Bermas, Ruby Mae O. Penolio, Adrian G. Saballero

which is entitled:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

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 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
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8. The parties acknowledge that any technology, product or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
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 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
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14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR POLYTECHNIC COLLEGES


NOEL V. PAGUIO JR.
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: May 21, 2025

DATE: May 21, 2025

APPENDIX G

JOINT AFFIDAVIT OF UNDERTAKING

Appendix Y. Joint Affidavit of Undertaking (Plagiarism)

Joint Affidavit of Undertaking

We, (1) NIÑO JAYBEE R. BERMAS, of legal age, single/ married and a resident of KINALABASAHAN BULA, CAMARINES SUR, (2) RUBY MAE O. PENOLIO, of legal age, single/ married and a resident of LOURDES YOUNG NABUA, CAMARINES SUR, (3) ADRIAN G. SABALLERO, of legal age, single/ married and a resident of SAN NICOLAS, IRIGA CITY, after having been sworn to in accordance with law, do hereby take oath and state :

- (i) That, we are officially enrolled for the thesis/capstone project on the topic titled INTEGRATION OF ROBOTIC PROCESS AUTOMATION (RPA) IN THE ACADEMIC OPERATIONAL TASKS OF THE COLLEGE OF COMPUTER STUDIES (CCS): A MODULE AUTOMATING THE PROCEDURE FOR GENERATION OF WORKLOAD SCHEDULE OF FACULTY in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.
- (ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN INFORMATION SYSTEMS Degree are original and our own work, and is not plagiarized.
- (iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.
- (iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.
- (v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this NOV 17 2025 day of 2025 in _____, Philippines.

(1) NIÑO JAYBEE R. BERMAS

Affiant

(2) RUBY MAE O. PENOLIO

Affiant

(3) ADRIAN G. SABALLERO

Affiant

NOV 17 2025

SUBSCRIBED AND SWORN TO before me this _____ day of _____ at _____, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. _____
Page No. _____
Book No. _____
Series of 202 _____

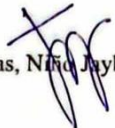
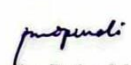

INAH SOPHIA T. CABALTERA
Public Attorney
Pursuant to RA 9408

APPENDIX H

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	BOSKOOOP
Course Code	IS 3212
CaPSA	ICHELE F. BALUIS, MSIT

Name and Signature	Project Role	Email address / mobile#(s)
 Saballero, Adrian G.	Project Head	adsaballero@my.cspc.edu.ph 09919459279
 Bermas, Nino Jaybee R.	Information Manager	jabermas@my.cspc.edu.ph 09955994663
 Penolio, Ruby Mae O.	Manuscript Writer	rmpenolio@my.cspc.edu.ph 09513919040

APPENDIX I

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Team Alias: Boskoop
Proponents/Researchers:

1) Adrian G. Saballero
2) Niño Jaybee R. Bermas
3) Ruby Mae O. Penolio

Proposed Thesis/ Capstone Project Title:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

Submitted by:  <u>SABALLERO ADRIAN G.</u> (Signature of Project Head over printed name) Date: <u>FEBRUARY 14 2025</u>	Noted:  <u>KHELLE F. BALUIS</u> (Signature of CAPSA over printed name) Date: <u>2/18/25</u>
Recommending Approval:  <u>SHEILA DONA S. MATA-APAZMIR</u> (Signature of CAPA over printed name) Date: <u>FEBRUARY 14, 2024</u>	Approved:  <u>ROSEL O. ONESA</u> OIC Dean, CCS Date: <u>2/18/25</u>

APPENDIX J

CAPSTONE HEARING FORM

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: May 19, 2025 ☒ Title Proposal ☐ Pre-Oral ☐ Final Oral

Date of Hearing: May 21, 2025 Time: 3:00-5:00pm Venue: AIRCODE

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College
of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

Proponent/s:

Adrian G. Saballero

Ruby Mae O. Penolio

Niño Jaybee R. Bermas

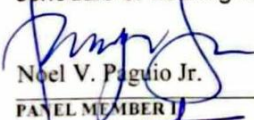
Recommended by:


Ms. Shiela Dora S. Manlapaz

Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]


Noel V. Paguio Jr.
PANEL MEMBER 1


Jonuel Rey Colle
PANEL MEMBER 2

Brenda D. Benosa, MIS
PANEL CHAIR

NOTED BY:


ICHELLE F. BALUIS
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC- DEAN, CCS

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: October 17, 2025 ☐ Title Proposal ☐ Pre-Oral ☐ Final Oral
Date of Hearing: October 21, 2025 Time: 3:00- 5:00pm Venue: Open Laboratory

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College
of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

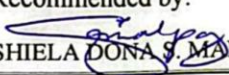
Proponent/s:

Adrian G. Saballero

Ruby Mae O. Penolio

Niño Jaybee R. Bermas

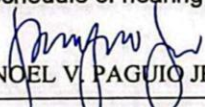
Recommended by:


SHIELA DONA S. MANLAPAZ, MIT

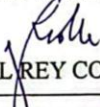
Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]


NOEL V. PAGUIO JR.

PANEL MEMBER 1


JONUEL REY COLLE, MIS

PANEL MEMBER 2


BRENDA D. BENOSA, MIS

PANEL CHAIR

NOTED BY:


ICHELLE F. BALUIS, DIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC- DEAN, CCS

THESIS/ CAPSTONE PROJECT HEARING FORM

Date filed: _____ [] Title Proposal [] Pre-Oral [x] Final Oral
Date of Hearing: November 26, 2025 Time: 1:00-3:00 PM Venue: AIRCODE

Department: COLLEGE OF COMPUTER STUDIES

Research Title:

Integration of Robotic Process Automation (RPA) in the Academic Operational Tasks of the College
of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule
of Faculty

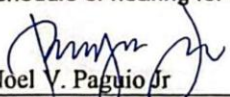
Proponent/s: Niño Jaybee R. Bermas Adrian G. Saballero
Ruby Mae O. Penolio

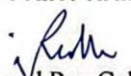
Recommended by:


Shiela Dona S. Manlapaz, MIT
Thesis/ Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]


Noel V. Pagnio Jr
PANEL MEMBER 1
Brenda D. Benosa, MIS
PANEL CHAIR


Jonuel Rey Colle
PANEL MEMBER 2

NOTED BY:


ICHELLE F. BALUIS, MIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
DEAN, College of Computer Studies

APPENDIX K

PANEL RSC

Title : Integration of Robotic Process Automation (RPA) in the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

Alias : Team Boskoop

Date : May 21, 2025, 3:00 PM | / | TD | | POD | | FOD

Secretary : Justine D. Labios

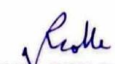
MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
1	1	1.1 Project Context Provide basis/reference	Added citation
1	2	- Problem basis - Module scale (education sector)	Revised the project context by removing general topics allowing to focus more on RPA and education sector
1	3	1.2 Purpose and Description of the Project - Please revise according to the needed content in the purpose and description of the project according to format. - discussion - Add uniqueness	Revised the construction by discussing the purpose and added a brief description of the context of the study.
1	4	2. Kindly replace the word 'process' into 'tasks' 1.4 Significance of the Study Rearrange	Replaced the word with 'tasks' Significance of the study is rearranged.
1	5	1.5 Scope and Limitations of the Project "Software Robot"	Added the term software robot and was discussed in the limitations.
1	6	- Please add 'The' before the word 'efficiency' - Include Software Robots (process task)	'The' was added to the word. Added 'Process', 'Task', 'Software Robots' and were discussed in the context of the study

2	11	2.1.2 Please add the word 'process' involved in 'Implementing automation in the Education Sector.'	'Process' was added in the sub-topic title of RRL.
2	14	2.1.4 Kindly replace the word 'Implementing' into 'Implementation'	Word replaced.
3	26	Automatability Assessment	Added Multi-Criteria Scoring Model
3	27	3.3.1 Data sources - Add data sources Table 1 - Kindly replace the word 'Checklist' into 'Journal' - Automatability Score Sheet	Data Sources were discussed and added. Word replaced. 'Automatability Score Sheet' added.
3	28	2. - Kindly replace the word 'Checklist' into 'Journal' 3. - Kindly replace the term 'Test Cases' into 'Simulation'	Replaced the word 'checklist' and 'journal'. revised the meaning.
3	30		-
3	31	Please add 'The' before the word 'efficiency'	The 'the' word was added.
3	32	Notes: Follow ACM citation format.	Format was double checked.

NOTED BY:


BRENDA D. BENOSA, MIS
 PANEL, CHAIRMAN


NOEL V. PAGUIO JR.
 PANEL MEMBER 1


JONUEL REY COLLE
 PANEL MEMBER 2

Title : **Integration of Robotic Process Automation (RPA) in the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty**

Alias : Team Boskoop

Date : October 22, 2025 | | TD | / | POD | | FOD

Secretary : Jerlyn Baluis

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
1	1,2	1.1 Project Context - Stick to 3-5 sentences per paragraph. <i>Applies to all chapters</i> - Please add more support to the need of your study. Elaborate discussion on the known observed problems that can be resolved through automation.	Pages 1-3 Paragraphs were guided accordingly. Sentences were revised and added studies to support coherent discussion.
1	4	1.4 Significance of the Study - Please revise. Make it more personal, how this automation can exactly help the faculty.	Pages 3-4 Revised the section by elaborating more on the importance of the study to each sector.
1	4	1.5 Scope and Limitations of - 1 paragraph for the scope; 1 paragraph for the limitations. Add SO's. - Add also limit on your respondents/participants and other parts of your methods.	Page 4-5 Added SO's and Limitations to the discussion.
1	5,6	1.6 Project Dictionary - Please operationally define the terms in the context of your study. - <i>Efficiency</i> : Indicate that you are meaning time saved.	Page 5-6 Reconstructed the terms based on the context of the study
2	10	Review of Related Literature - Please make your discussion coherent. - Make sure of a smooth transition from 1 par to the next. Do not just enumerate. <i>(Applies to all topics.)</i>	Pages 10-20 Discussion were reviewed making sure that the context is not only enumerated rather discussed.

2	18	2.2 Synthesis of the State-of-the-Art - Present how these research/RL relate to your study (similarities/differences). Do this for each topic.	Page 20-22 Identified similarities in the context of the study and added the differences.
2	20	2.3 Gap Bridge by the Study - Identify gaps based on the synthesized RL. What certain conditions/objectives that were not met/covered by existing literature? Present how your study was able to address them all.	Page 22-23 Revised the context of the section by identifying the gaps based on the synthesized RL.
3	27	3.1 Business Process Management Framework - <i>Process Discovery</i> : As-Is model. - <i>Process Analysis</i> : Problems. - <i>Process Redesign</i> : To-Be - <i>Process Implementation</i> : Automation proper. - <i>Process Monitoring</i> : Evaluation	Page 29 Frameworks were discussed according to the application of the processes.
3	29	3.3.1 Data Sources - Identify your data sources (faculty, deans office staff). Then mention how many of them. - Also did you use documents, etc? These are considered as data sources. - Have a subsection for respondents & sampling techniques.	Specified data sources such as documents that were found significant in the study. Page 32-33 Sampling Techniques and the Respondents which were in Data Analysis Procedure were moved in this section.
3	30	3.3.2 Instrument - Please discuss in the context of your study. After the conceptual definition, describe the instrument based on your study (TD RSC). - 'Simulation Journal' - Add an inspection journal for the accuracy test.	Page 35 Instruments were rephrased based on the context of the study. 'Journal' was added. Inspection Journal added
3	30	3.3.3 Data Gathering Procedure - Add simulation inspection.	Page 36-40 , Simulation and Value-Complexity Matrix added
3	31	• <i>BPM Lifecycle-Process Analysis</i> - Change this to the name of the model. • <i>Table 2 Multi-Criteria Scoring Model</i> - Provide f'la and discussion on how this	Page 37 • renamed to Multi-Criteria Scoring Model • Discussion were added as a conclusion connecting to Range Scale

		will be populated and calculated.	
3	32	- Table 3 Range Scale for Degree of Automation - Mention references and how intervals were calculated.	Page 39 Range Scale for Degree of Automation were updated, citation added, intervals were discussed after table.
3	32	3.3.2 Data Analysis Procedure Separate section sampling technique.	Page 41 Sampling Technique was put in Data Sources
3	34	- Accuracy & Efficiency - Please discuss how these formulas will be used in the context of your study.	Page 42 Added a short discussion before the Efficiency and Accuracy is presented
4	38	4.1.1 Existing Process Schedule Generation 'Workload Process Generation'	Page 46 Name changed.
4	39	- Figure 3 Process Overview of Schedule Generation Procedure - Creates Workload Schedule, shade this for emphasis.	Page 47 Shaded to yellow.
4	41	4.2 Process Discovery (As-Is) Discussion per task (trigger first), with subsection open SLN. 1 to 2 sentences per task to be able to compare easily with the to-be model.	Page 49 Tasks in As-Is were discussed.
4	41	4.2.1 Visualization of Schedule Generation 'As-Is Workload Generation Process' - Discuss per task.	Page 49 Name Changed. Page 50-52 Tasks were discussed.
4	42	4.3 Process Analysis - Identify the bottlenecks / needed improvements. Discuss where the problem's coming from based on the interview conducted with faculty. - Add thematic analysis as your method. - To-Be model, discuss per task.	Page 53 Bottlenecks were discussed in summary of each analysis. The use of Thematic Analysis discussed P. 53 . Interview-based results were also added. P. 58 To-Be model tasks were discussed, P. 61 .

		(If overload is not included, state that in your limitations.)	The Value-Complexity Matrix is added to support these findings, P. 59.
4	43,44	4.3.1 Automatability Assessment using Multi-Criteria Scoring Model - Include all the defined tasks. • <i>Table 5 Multi-Criteria Score Model</i> - Double check scoring.	Page 50 All tasks were included and were discussed each, P. 55-58. Scoring updated. P. 54
4	46	4.4.1 Revamp Model of Schedule Generation - 'To-be Model Workload Generation'	Page 60 Name changed
4	48	4.4.2 Technical (To-Be Process) Model - Per task discussion please.	Page 64-69 Tasks were discussed based on how the software robot works.
4	All	Clear image of your figures please. Figure 5-7	Images were updated for a clearer visualization of the diagram and texts
4	49	4.5.1 Hardware and Software Tool Specifications - Hardware specifications, consider not just running the RPA and Excel but other apps as well. - Refer to the recommended requirements, not the minimum ones. - Add monitor specs.	Page 70 Section revised according to the specifications needed to perform RPA with Virtual machine. Monitor or Screen Resolution added, P. 71.
4	51,55	4.5.2 Integration of Automation using Microsoft PAD - Add the start of the automation. - Add a sample sent and generated workload to faculty. - Redact faculty names or use dummy names on your screenshots. <i>True to all images.</i>	Page 73 - Added the start of automation - Added a sample sent in the <i>Distribution of File through Email</i>, P. 76 - All images were redacted in solid color: yellow.
4	56	4.6.1 Non-Probability Approach For sampling technique, use stratified sampling with 30 target participants. Include in your limitation that the basis of the data collected were the availability and willingness of participants.	Stratified sampling and sampling techniques were moved and discussed in chapter 3. P. 31 Limitations were also included based on the willingness of participants.

		<i>Update your chapter 5 based on the result.</i> • <i>Table 8 Distribution of Respondents</i> - Move to chapter 3.	Chapter 5 updated. • Table moved in chapter 3
4	57	4.6.2 Efficiency Results of Automation - Introduce the tables.	Page 81 4.6.1 Efficiency Results of Automation - Tables introduced
4	All	After the last phase, include a section for 'other insights'.	<i>Insights were added based on the acquired results</i>
4	58	4.6.2 Accuracy and Efficiency Results of Automation. - Operational definition of 'accuracy' based on the generated content to justify accuracy.	Operational Definition of Accuracy added, <i>P. 80.</i> 4.6.2 Accuracy Results of Automation, P. 82.
4	All	Make the diagrams readable.	<i>Texts made bigger</i>
5	62	5.1 Summary - Incomplete. Please provide complete content of your summary.	Revised the discussion by adding the methodologies and framework used in the study. <i>P. 88.</i>
5	63	5.2 Findings - Describe first the existing process then discuss the results of the automatability assessment. How many are low, mid, high in percentage then mention the significant tasks. - <i>Findings 1:</i> Mention all the methods used. - <i>Findings 2:</i> Mention the to-be model. - <i>Findings 3:</i> Mention the methods used.	- Existing process were put to discuss the task and results of automatability assessment. - Findings 1: mentioned the methods that were used - Findings 2: To-Be Model mentioned - Findings 3: Methods used were mentioned.
5	64	5.3 Conclusions - Update accordingly.	Conclusions were updated based from the findings. <i>P. 91</i>

CAPSTONE PROJECT OUTPUT	
RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
After automation for one faculty member, save the file, then delete the sheet before automating for the next faculty member. This helps reduce file size and ensures each faculty member sees only their own workload.	Task flow for deletion of 'sheets' were added before saving each file.
After automation, there should be a summary report indicating - successful entries, - skipped entries and conflicts. - names/email addresses where email failed to send and the reason.	Added a Summary of Report for - successful entries - Conflicts - Failure to send were added
Perform a simulated email test. - Use email routing to create dummy emails where automation will send the file, but received in one inbox. Present the received email, it should have the same time stamp. - During FOD simulate using the CCS email and send to 70 faculty emails.	- Subscribed a 1 year domain to Hostinger to produce dummy emails using cloudflare -
Autorun RPA on a virtual box to allow multitasking. Provide .ova file.	Oracle VirtualBox were used to add '.OVA' File for easier execution of the automation.
Check your pop-up message, there are incorrect spellings.	'Auotomation' mis-spelled word corrected.

NOTED BY:


BRENDA D. BENOSA, MIS
 PANEL, CHAIRMAN


NOEL V. PAGIO JR.
 PANEL, MEMBER 1


JONUEL REY COLLE
 PANEL, MEMBER 2

Title : **Integration of Robotic Process Automation (RPA) in the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty**

Alias : Team Boskoop

Date : December 3, 2025 | | TD | | POD | / | FOD

Secretary : Cecil D. Dagon

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN
1	4	1.4 Significance of the Study - Please revise the discussion of faculty. - Change the word 'college staff by' to 'faculty'.	Page 4 - Significance to Faculty were revised. (Significance to the College of Computer Studies) Word changed.
	4	1.5 Scope and Limitations of the Project - Please restructure and check your methods. - Who are the participants? Faculty? etc. and revise for clarity.	Pages 4-5 - Restructured by mentioning the methods from Chapter 3. - Mentioned 'Faculty' and were revised for clarity.
	6	1.6 Project Dictionary - Change the word 'system' to 'technology'. - With vs. without RPA integration	(RPA definition: In this study, RPA is a <i>technology</i>..) Word Changed. P. 5 - Time-efficiency definition revised. P. 6
3	34	Table 1: Target Sample Size - Where are the other respondents? Include them in the table. - Recompute based on the acceptable values. - Change '13%' to '5% acceptable margin error'	Page 34 - Added Dean Staff - Table Recomputed - Margin of Error updated
	38	Add automatic Mean on your discussion which will be the basis of interpretation.	Added Arithmetic Mean in the discussion. P. 38

	41	3.3.2 Data Analysis Procedure - Revise RPA help redesign the TO-BE or TO- BE model guided RPA. - Remove the other part of the sentences.	Page 41 - Section revised. - Part of the sentence was removed.
	42-43	Arithmetic Mean not weighted mean	Word and definition changed. P. 42
	44-45	ACM format	ACM updated
4		- No PILLS discussion - Remove the word 'process' in 4.1.1	- Topic flow were discussed in PILLS format - Word removed. P. 46
	49	Change the word 'schedule' to 'workload'.	Name changed. P. 49
	54	Add column before the task for the identifier eg. A,B etc.	Column added and named as 'Task' .P. 54
	55	1st statements	Moved as 1st statement. P. 54
	60	Mention which of this task was automated	Revised for clarity. P. 60
	61	- Please be consistent with your term - Change the word 'schedule' to 'workload' in figure 6	Terms updated - Figure 6 Schedule were changed. P. 61
	64	4.4.2 Technical (To-Be Process) Model Which process? Workload generation?	'of Workload Generation' Words Added. P. 64
	82	Mention the conversion (in minutes).	Minutes were added. P. 82
	87	ACM format	
5	90	Overall results?	Added the time saved. P. 90

CAPSTONE PROJECT OUTPUT	
RECOMMENDATIONS, SUGGESTIONS AND COMMENTS (RSC)	ACTION TAKEN

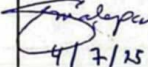
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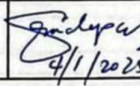

BRENDA D. BENOSA, MIS
PANEL, CHAIRMAN


NOEL V. PAGUIO JR.
PANEL, MEMBER 1


JONUEL REY COLLE
PANEL, MEMBER 2

CONSULTATION LOGS FORM

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:	Boskoop						
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Chapter 1	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	4/7/25						 4/7/25
	Remarks:						
	improve RPA in diff. software & method use						
	in RPA 1.0						

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:	Boskoop						
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Chapter 2	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	4/1/2025						 4/1/2025
	Remarks:						
	Revise Arrangement & Construction of Chapter 2						

CONSULTATION LOGS FORM

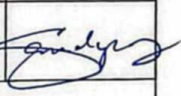
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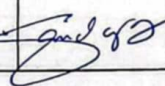
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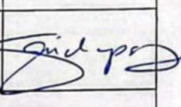
CONSULTATION LOGS FORM

Capstone Project Title:		Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:		Boskoop						
Proponents:		Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Chapter 4 - 5	Date of Consultation						Project Head's Signature	CAPA/Consultant / Grammarian Signature
	12/27/25							
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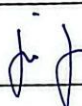
CONSULTATION LOGS FORM

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:	Boskoop						
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Chapter 4	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	10-15-25						
	Remarks:						
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	Chapter 3 make it total enumeration						
	But on						
	Chapter 4 provide the actual						
	survey						

CONSULTATION LOGS FORM

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:	Boskoop						
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Chapter 1 - 5	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	10-17-25						
	Remarks:						
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	of our 5%						

CONSULTATION LOGS FORM

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty							
Alias:	Boskoop							
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio							
Project Output	Date of Consultation						Project Head's Signature	CAPA/Consultant / Grammarian Signature
	11/19/25							
	Remarks:							
	Good to go!							
	Recommended for FOD!							

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty							
Alias:	Boskoop							
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio							
Chapter 4 - 5	Date of Consultation						Project Head's Signature	CAPA/Consultant / Grammarian Signature
	10/1/25						[Signature]	[Signature]
	Remarks:							
	- Recommended for POD!							

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
Alias:	Boskoop						
Proponents:	Adrian G. Saballero, Niño Jaybee R. Bermas, Ruby Mae O. Penolio						
Project Output	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	12/15/25						
	Remarks:						
	Recommended for Bluebook!						

CONSULTATION LOGS FORM

Capstone Project Title:	Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty						
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Manuscript	Date of Consultation					Project Head's Signature	CAPA/Consultant / Grammarian Signature
	12/14/25						
	Remarks:						
	PROPOSED FOR BLUEBOOK.						

APPENDIX M

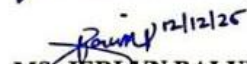
SECRETARY CERTIFICATE

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project entitled **“Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty”** prepared and submitted by Niño Jaybee R. Bermas, Ruby Mac O. Penolio, and Adrian G. Saballero and that the same have not been amended, modified or obliterated.

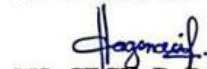
Signed:



MR. JUSTINE D. LABIOS
Title Defense Secretary



MS. JERLYN BALUIS
Pre- Oral Defense Secretary



MS. CECIL D. DAGON
Final Defense Secretary

Date signed: 12/12/25

APPENDIX N
GRAMMARIAN CERTIFICATE

This is to certify that the undersigned has reviewed and went through all the pages of the capstone project entitled **“Integration of Robotic Process Automation (RPA) 1.0 In the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty”** as against the set of structural rules that govern research writing in accord with the composition of *sentences, phrases, and words* in the English language.

Signed:


MR. KHARL VINCENT C. REQUINALA
Grammarian

Date signed: 11/26/25

APPENDIX O
CERTIFICATE OF TRANSFER

CERTIFICATE OF TRANSFER

This is to certify that the

**"INTEGRATION OF ROBOTIC PROCESS
AUTOMATION (RPA) 1.0 IN THE ACADEMIC
OPERATIONAL TASKS OF THE COLLEGE OF
COMPUTER STUDIES (CCS): A MODULE
AUTOMATING THE PROCEDURE FOR GENERATION
OF WORKLOAD SCHEDULE OF FACULTY"**

developed by *Niño Jaybee Bermas, Adrian Saballero
and Ruby Mae Penolio*, as their undergraduate
capstone project was transferred to the Dean's Staff of
the College of Computer Studies of the Camarines Sur
Polytechnic Colleges.

This certificate is issued this 19th day of January, 2026,
at the CCS Office of the Dean, for all valid purposes it
may serve.


ROSEL O. ONESA, MIT
OIC, Dean of CCS

APPENDIX P

ACM FORMAT

Integration of Robotic Process Automation (RPA) 1.0 in the Academic Operational Tasks of the College of Computer Studies (CCS): A Module Automating the Procedure for Generation of Workload Schedule of Faculty

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Employees in modern workplaces increasingly rely on technology to manage operational processes yet requires proper planning and implementation to remain efficient. In the College of Computer Studies (CCS) at Camarines Sur Polytechnic Colleges, several long-standing procedures particularly in the manual preparation of documents have become repetitive, and time-consuming. This study, *The Integration of Robotic Process Automation (RPA) to the Academic Operational Tasks of the College of Computer Studies*, aims to enhance the *Generation of Workload Schedule of the Faculty* by assessing its automation potential and effectiveness. A multi-method approach was used to meet the study's objectives. Qualitative methods, including interviews and observations, identified workflow patterns and faculty concerns, while quantitative methods provided numerical interpretations through scoring formulas, data inspection, and simulation. Guided by the Business Process Management (BPM) Framework, the researchers (1) analyzed the existing workflow mapped in As-Is BPMN model, (2) evaluated automatability using Multi-Criteria Scoring Model and Value-Complexity Matrix, and (3) developed a To-Be Model. Nineteen tasks were identified, with seventeen (89.47%) classified as highly automatable and two (10.53%) as medium. Four tasks were deemed highly complex due to complex plotting requirements. The redesigned automated workflow implemented Microsoft Power Automate Desktop in VirtualBox showing a reduction in processing time from an average of 4 minutes and 60 seconds manually to 37 seconds with automation, achieving 100% accuracy across ten generated documents. The study concludes that RPA 1.0 significantly improves efficiency and reliability and recommends exploring more advanced RPA versions, to other departments, and refining workflows for future research

CCS CONCEPTS • Applied computing • Enterprise computing • Business process management • Business process management systems

Additional Keywords: Automation, Software Robot, Workload Schedule, Manual Practices

1 INTRODUCTION

Technology has changed how daily task are done especially in workplaces that handle repetitive tasks. In educational institutions, technology helps reduce manual works, save time, and improve efficiency [2]. However,

studies show that stress and burnout in work can still happen when task crash even if they seem small at first [5, 10]. The College of Computer Studies (CCS) at Camarines Sur Polytechnic Colleges (CSPC) has already use Robotic Process Automation (RPA) in operational task such as grade related procedure, validation of student enrollment procedures, and adviser reportorial requirements, which helped reduce processing time and workload [3, 4]. Despite this progress, the process of creating workload schedule of faculty is still done manually which requires a lot of time and can cause delays and work-related concern. This study aims to improve the current process of faculty workload schedule generation using Robotic Process Automation (RPA1.0). The study examines the existing process, identifies the task that can be automated, and evaluate the accuracy and efficiency of the implemented automation. The scope of this study focuses only on plotting of weekly schedule based on the Subject Load Notice (SLN) file using Power Automate Desktop (PAD). Task that requires human interaction such as overload, quasi teaching, research assignment, etc. are not included since they still need human interaction. This study is important because it can help faculty to save time and focus more on teaching and other important task. It also helps researchers and future researchers to better understand how RPA can be used in administrative task within academic institutions.

2 BACKGROUND

Educational institutions like the College of Computer Studies at the Camarines Sur Polytechnic Colleges operate technology-driven approaches to its operations. In its previous years, the college has effectively deployed Robotic Process Automation (RPA) that were found efficiently functional in the college. Time durations were cut down in automating procedures that require heavy uptime work like encoding grades and obtaining class lists. In spite of this milestone, the college still faces burdens on its operational tasks particularly in creating workload schedules. Although the role of technology were considered to ease manual operated tasks, the tasks of plotting these schedules in each faculty show significant burden to the functionality of both the administrative office and the instructors. Studies show that timetable scheduling requires significant time and meticulous planning [6]. In the current set-up of the college, the tasks of plotting each schedule were passed down to each instructor to process documents which can result in inefficiencies and work complaints. Considering that the process of generating a workload schedule of each faculty amounts to a huge manual-processing time, utilizing the use of automation technology, according to Charles et. al, offers a strategic move to ease and efficiently optimize workflow of employee tasks [5]. Hence, a move to integrate RPA shows relevancy in terms to effectively managing a complex work and satisfaction to work-related tasks [8]. With automation technology that interacts well with applications and executes 'flows' the same as how a user typically performs a task repetitively. This allowed users to deploy robots to interact within applications thus supporting their tasks in certain processes that are still needed for human interaction [1, 7]. As a result, the integration of RPA solutions posed a great opportunity to refine manually processed tasks and re-digitalized repeated action of manually driven activities. The application of workload distribution systems provides a noticeable gap leaving a limited discussion on technical implementation of automation related to faculty workload-schedule plotting. Bédard et al. [2024], were mainly concerned with determining the value but lacks application of RPA. Furthermore, many automation tools were introduced particularly using heavy-programming skills like Python but lacks discussion to underlying tasks that is tailored in an institutional setting. By developing an RPA system focused on structured-based approaches capable of applying automation in plotting faculty workloads, this study will adapt the use of BPM framework as it is believed to improve automation's overall effectiveness and further align them

with the organizational goals of the College. In addition, this study bridges conceptual insight from past studies with a concrete application of *automating the procedure of workload schedule* using Microsoft Power Automate Desktop.

3 METHODOLOGY

3.1 Business Process Management (BPM) Lifecycle Framework

In this study, the BPM framework is adopted to effectively analyze, model and evaluate the process of workload schedule generation within the College of Computer Studies (CCS). This framework provided the researchers with a clear and complete perspective of existing workflows, allowing them to identify operational bottlenecks and develop a more efficient solutions. This framework consists of 6 phases, process identification, process discovery, process analysis, process redesign, process implementation, process monitoring.

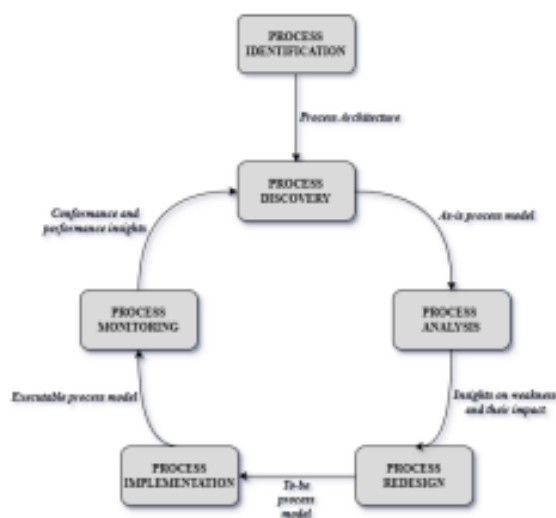


Figure 1: Business Process Management Lifecycle

During the process identification, the researchers identified and defined the steps involved in generating faculty workload schedule by conducting semi-structured interview and direct observations to understand how the process is currently performed in College of Computer Studies (CCS). Process Discovery followed by collecting detailed information about the existing or the As-Is model, where each task involved in workload schedule generation was documented well to clearly show how the process is done. Process Analysis was then conducted to examine the collected data and identify automatable tasks using automatability assessment and scoring sheet. During Process Redesign, To-Be process model was developed using BPMN, integrating RPA 1.0 through Microsoft Power Automate Desktop to automate the task identified in Process Analysis and improve overall efficiency. Process Implementation involved executing the redesigned automated workflow by extracting

data from the Subject Load Notice (SLN) file, generating faculty workload schedule, and distributing the generated workload schedule to faculty using Microsoft Power Automate Desktop. Finally, Process Monitoring focused on evaluating the automated process using predefined metrics to measure the accuracy and efficiency.

3.2 Research Design

This study employed a multi-method design to effectively achieve its objectives. Both qualitative and quantitative approaches were used to gain a comprehensive understanding of the research problem. Qualitative methods, such as interviews, observation journals, and document analysis, allowed the researchers to interpret processes and contexts related to the study. Quantitative methods, including rating scales, Likert scales, and score-based rubrics, were used to measure and validate numerical data for clearer analysis.

The combination of these approaches strengthened the validity and reliability of the findings by integrating contextual insights with measurable evidence. Rather than relying on a single method, the quali-quant approach complemented the strengths of both. This integrated design aligned with the Business Process Management (BPM) Framework, ensuring a systematic analysis, redesign, and evaluation of the faculty workload scheduling process and providing a reliable and clearer assessment of the impact of Robotic Process Automation.

3.3 Research Method

3.3.1 Data Sources

To achieve the goals of the study, the implementation of qualitative and quantitative approach were used as an instrument for data gathering procedures. The primary and secondary sources were identified in this section.

Faculty. Consisting of 63 in total, the faculty provided first-hand knowledge of how they manually plot their subject load notice.

Dean's staff. This source offered insights into procedural and managerial aspects of workload scheduling, helping the researchers understand the overall process and the significance of the faculty's subject load.

Subject Load Notice (SLN) Document. This document provided accurate information such as faculty names, subject load, designations, and academic year, which served as a reference for plotting schedules.

Workload Schedule Template. Is a document containing details such as time, day, room, subject code, program year, section, current semester, and faculty assignments, and is an ISO-registered document used for manual schedule plotting.

3.3.2 Instrument

The study made use of several data-gathering tools to ensure that the results were completely reliable and legitimate. Semi-structured interviews were conducted to gather detailed information and to better understand the documents and procedures involved in faculty workload plotting. An observation journal was also used to directly record how manual tasks were performed, including the challenges encountered and how a faculty and dean's staff interacted with software applications during the process.

A simulation journal was used to compare the time efficiency of manual practices with the simulated automated solution, helping identify the time saved using RPA. While, an inspection journal was utilized to assess the accuracy of the automated outputs by checking important details such as schedules, names, and course information, ensuring that the automation produced the expected and correct results.

3.3.3 Data Gathering Procedure

The study used several methods to understand and improve the faculty workload plotting process. Interviews with faculty members and the dean's staff were conducted to gain insights into current procedures and identify areas for improvement. Observations were also performed to examine how schedules are prepared and to understand user-system interactions. Simulations were then used to evaluate the effectiveness of the RPA solution by comparing manual and automated scheduling times. Inspections followed to ensure the accuracy and consistency of the automated outputs. Lastly, an automatability assessment, supported by the Value-Complexity Matrix, was conducted to identify tasks suitable for automation and guide the development of the proposed RPA system.

3.3.4 Data Analysis Procedure

The evaluation of the faculty workload schedule focused on supporting improvement and ensuring accurate and fair analysis of the collected data. Two main measures were used: accuracy and efficiency, both based on data gathered through interviews and observations. These measures assessed whether the integration of RPA resulted in meaningful improvements compared to existing practices. Accuracy measured the correctness of the outputs generated by the software robot, including schedule details and information from the Subject Load Notice (SLN). Efficiency measured the time saved by comparing manual task completion with automated execution. The Arithmetic Mean was used as the basis for interpreting results, providing a clear and consistent evaluation of the effectiveness of RPA in the workload scheduling process.

4 RESULTS AND DISCUSSION

4.1 Process Identification

The researchers were able to gather first-hand information by conducting semi-structured interviews to the faculty members and the dean's staff. These gathered data were then served as the basis for the interpretation of each process specifically graphed as 'As-Is' Model using Business Process Model Notation (BPMN).

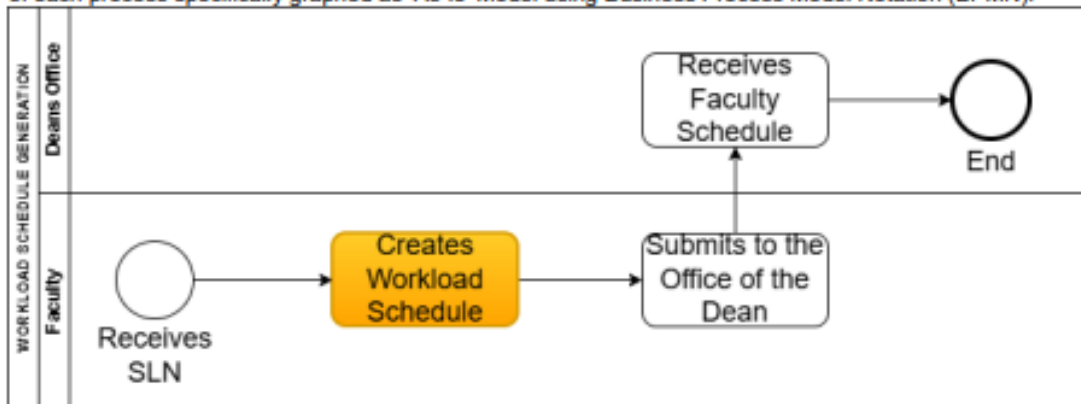


Figure 2: Process Overview of Schedule Generation Procedure

The current process revealed an inefficient manual workload scheduling process and is slowed down by human dependency and potential repetitive tasks. The BPMN diagram provides insights into workload scheduling workflow. The findings highlight the benefits of adopting automation to enhance efficiency, accuracy and timelessness in creating the workload schedule. These tasks and workflow show that the college follows a structured process where faculty generates individual schedules.

4.2 Process Discovery

The 'As-Is' Process that were graphed revealed that the Subject Load Notice (SLN) is a crucial file that initiates the entire process. The process also exposes inefficiencies particularly in the plotting of each assigned schedule as potential errors especially in the inputs of time, date and other relevant information, cross-checking between apps, uncertain schedule, and unstructured repetition of tasks that often increase work burnout. The researchers also discovered that while the study aims to navigate the manual process of generating workload schedules by applying automation for lesser human intervention, the integration of RPA will still require human actions such as adding schedules not indicated in the SLN file.

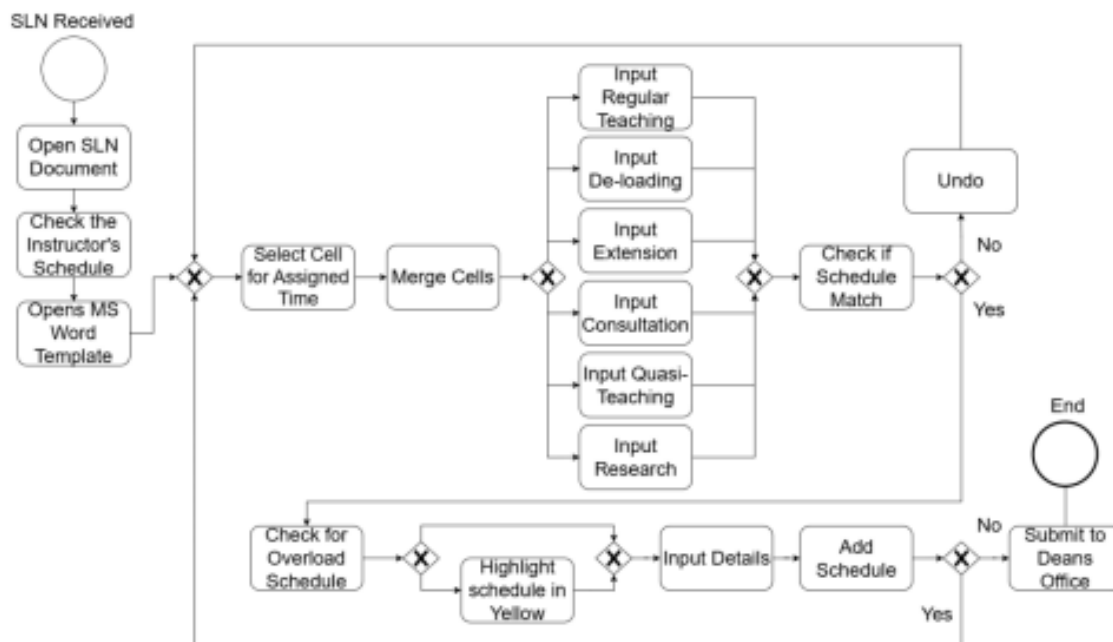


Figure 3: Descriptive As-Is Model Visualization of Workload Generation

The Subject Load Notice (SLN) plays a major role in Workload Schedule Generation process. Using As-Is model for the visualization of these tasks, it revealed that the steps in making Workload Schedule, shows an unstructured repetition which may reflect a huge amount of processing time. The researchers also discovered that while the study aims to navigate the manual process of generating workload schedules by applying

automation for lesser human intervention, the integration of RPA will still require human actions such as adding schedules not indicated in the SLN file.

4.3 Process Analysis

The identified tasks related to the plotting of the faculty schedules were assessed using multi-criteria scoring model and each task were evaluated in each criteria, the scores reflected were based on the level of automatability in a predefined degree of scales. Out of 26 tasks, 2 were marked as medium, while 24 were marked to be automatable. Another assessment were done to support and understand the ability of the automation to perform each tasks. Using Value-Complexity Matrix, the tasks were mapped out on the matrix to reveal the complexity of each task. The results revealed that the tasks A, and N along with Tasks E, F, G1, G2, H, J, K, L, M are plotted in the 1st quadrant which shows that these tasks offer high value and is crucial in the plotting process. These tasks indicate a top priority and can easily be automated due to their low complexity level. Tasks B, C, D, I, also display a low complexity level which can also be automated, but these tasks offer a lesser value. Meanwhile in the far-right quadrant, G3, G4, G5, and G6, are tasks that have the most value than any other tasks. Although these tasks indicate a high value in the plotting process, these tasks also offer higher complexity in the process which require complex variables that are difficult for automated systems to comprehend.

4.4 Process Redesign

Based on the assessments done during the analysis made by the researchers, the graphed 'As-Is' Process Model were optimized and named as 'To-Be' Process Model to develop a logically model of process integrated with automation technology.

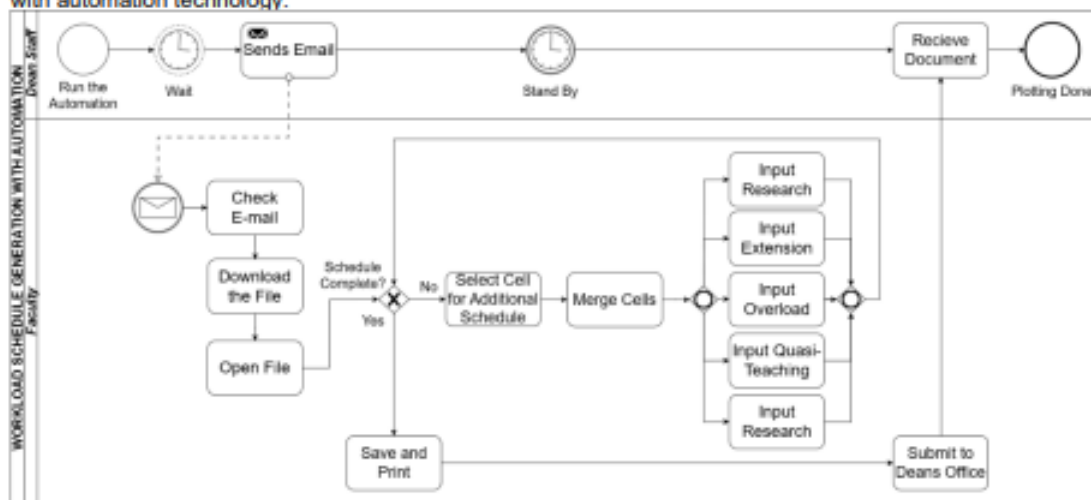


Figure 4: To-Be Process Model of Workload Generation

The logical flow of the To-Be process model were mapped out to effectively understand the structure. This remodeled process introduces a new efficient workflow reducing the repetition of tasks and errors. The transition

to RPA 1.0 technology were based on the previous 'As-Is' Process which were digitalized to consistently perform manually processed tasks and significantly improve processing of the document.

4.5 Process Implementation

The intended improvements were carried out to actual operations using the required hardware and software tools. Several specifications of both hardware and software were necessary to interact between the automation commands. Each command were tested that highlights areas for improvement and continuous fine-tuning of the workflow, adjustments to automation errors and enhancing system performance. Moreover, the use of virtual machine was added to strengthen the reliability of performing the automation workflow without the need to modify the automated-set activities when executed. Additionally, the Standard Operating Procedure (SOP) were generated to guide the user/staff in running the automation technology.

4.6 Process Monitoring

The monitoring of the process was conducted to evaluate the performance and effectiveness of automation input to the faculty workload schedules. This phase focused in two key metrics: the accuracy and efficiency, which display the interaction of the faculty to the automation.

4.6.1 Efficiency Results of Automation

The study used a table to clearly compare how long it takes to plot faculty workloads manually versus using an automated system. The table shows the time spent on each task and the average time, all measured in seconds, based on ten faculty members who plotted their schedules from the previous semester. Results indicate that manual plotting took about 2,998.1 seconds (nearly 50 minutes) in total, with each faculty member spending an average of around 299.8 seconds (almost 5 minutes) completing the task.

Table 1: Time Efficiency Evaluation Result

Faculty	Generated Outcome without RPA	Generated Outcome with RPA	Time Saved
Regular Faculty 1	496.8 seconds	43 seconds	453.8 seconds
Regular Faculty 2	621 seconds	37 seconds	584 seconds
Regular Faculty 3	249.6 seconds	31 seconds	218.6 seconds
Regular Faculty 4	435.6 seconds	34 seconds	401.6 seconds
Contract of Service 1	180 seconds	21 seconds	159 seconds
Contract of Service 2	143.4 seconds	34 seconds	109.4 seconds
Contract of Service 3	331.2 seconds	44 seconds	287.2 seconds
Contract of Service 4	237.5 seconds	34 seconds	203.5 seconds
Part-Timer 1	88.8 seconds	24 seconds	64.8 seconds
Part-Timer 1	214.2 seconds	25 seconds	203.5 seconds
Total	2,998.1 seconds	327 seconds	2,671.1 seconds
Average Time	299.8 seconds	32.7 seconds	267.1 seconds

The application of Robotic Process Automation (RPA) showed a significant faster processing time. The software robot were able to complete all schedules in just 327 seconds (about 5.5 minutes), averaging only 32.7 seconds per faculty member. This resulted in a total time savings of 2,671.1 seconds (around 45 minutes), equivalent to an 89% improvement in efficiency. As shown in Table 1, these findings demonstrate that automation greatly improves workflow efficiency, especially for repetitive tasks typically handled by faculty members.

4.6.2 Accuracy Results of Automation

The accuracy of the automated system was reviewed by the faculty to ensure that the faster results produced by automation were also correct and reliable. Accuracy was measured in a score of "1" that meant accurate and "0" for inaccurate. The evaluation focused on document format, as well as the correctness of the schedule in terms of day and time, all based on the information provided in the Subject Load Notice (SLN).

Table 2: Accuracy Inspection Data

Metric	Value
Total Samples	10
Correct Outcome/s	10
Incorrect Outcome/s	0
Accuracy	100
$\lfloor \text{Correct Outcomes} / \text{Total Samples} * 100 \rfloor$	

The results showed that the automated system achieved 100% accuracy for all ten faculty members. Feedback from regular faculty, instructors, part-time faculty, and contract-of-service instructors confirmed that the schedules generated by the automation were properly formatted, correctly assigned, and matched the details in the SLN. Some faculty suggested minor improvements, such as adding schedules that were not included in the source file, but these did not affect the overall accuracy of the system. Overall, the findings show that the RPA system consistently produced accurate workload schedules and proved to be a reliable tool for handling repetitive scheduling tasks.

5 CONCLUSION

Using Automatability Scoresheet, most tasks were revealed to be suitable for automation, while the Value-Complexity Matrix helped prioritize high-value tasks and identify those too complex for a software robot to perform. In the integration of RPA, a visual diagram named To-Be Process Model was modeled using BPMN to understand technicalities and the logically structured flow. A virtual machine (VirtualBox) was deployed for smooth implementation of the automation which helped demonstrate the automated solution to consistently produce workload schedules with 100% accuracy and reducing the overall process time by over 89%. Additionally, the flow control logic of the automation process may be improved by future researchers by adding the schedules not indicated in the SLN file. Other methodology other than Business Process Management (BPM) Methodology can be explored to further enhance scalability and efficiency while ensuring that it is aligned in the objectives and scope to their study. The integration of automation can be applied in a wider scope of the college or other department existing in the institution. Other automation tools can be used as this study highlighted RPA 1.0 as an effective and reliable tool for improving operational efficiency within the College of Computer Studies.

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APPENDIX Q

CERTIFICATE OF PLAGIARISM CHECK

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CERTIFICATION			
Date of Release:	January 20, 2026		
Submission ID:	23367:126541414		
Output Title:	INTEGRATION OF ROBOTIC PROCESS AUTOMATION (RPA) 1.0 IN THE ACADEMIC OPERATIONAL TASKS OF THE COLLEGE OF COMPUTER STUDIES (CCS): A MODULE AUTOMATING THE PROCEDURE FOR GENERATION OF WORKLOAD SCHEDULE OF FACULTY		
Author(s):	Adrian G. Saballero Niño Jaybee R. Bermas Ruby Mae O. Penolio		
Program:	Bachelor of Science in Information System		
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